| Results for SBCM Calibration for TOSS: NSLSII-INS-PRC-005 |
|---|
| Procedure Performed on 1/17/2025 by Anthony Caracappa and Mike Capotosto |
| Analysis Performed on 1/17/2025 by Anthony Caracappa |

| Test Equipment Used for Calibration | | |
|---|---|---|
| Oscilloscope | Manufacturer | Tektronix |
| | Model Number | MSO64B |
| | Serial Number | B027138 |
| | Calibration Expiration | 12/30/2025 |
| RF Source | Manufacturer | Rohde & Schwarz |
| | Model Number | SMB100B |
| | Serial Number | 102801 |
| | Calibration Expiration | 5/22/2026 |
| DMM1 | Manufacturer | Keithley |
| | Model Number | 2100 |
| | Serial Number | 8020356 |
| | Calibration Expiration | 3/20/2025 |
| DMM2 | Manufacturer | Keithley |
| | Model Number | 2100 |
| | Serial Number | 8020357 |
| | Calibration Expiration | 3/20/2025 |
| Python Scripts | Bandwidth Scan | BandwidthScan.py |
| | Characteristic Curve | CharacteristicCurve.py |
| | Beam Scan | BeamScan.py |
| | Report Generation | FunctionalTest.py |
| | DCCT Report | ProcessSBCM.py |

| Preliminary Settings Adjustments | | |
|---|---|---|
| Adjustments made on 1/17/2025 by the CE: Anthony Caracappa | | |
| Adjustment | Set To: | Status |
| Rectifier Offset A | 5.9mV | PASS |
| Rectifier Offset B | 34.1mV | PASS |
| Downconverted Frequency A | 1.8MHz | PASS |
| Downconverted Frequency B | 1.81MHz | PASS |
| Measurement | Reading | Status |
| Downconverted Frequency A Level | 272.0mVpp | PASS |
| Downconverted Frequency B Level | 252.0mVpp | PASS |

| Power Supply | |
|---|---|
| Actual Voltage (V) | Measured Voltage (V) |
| +12.0 V | 11.95 |
| -12.0 V | -11.85 |
| +5.0 V | 5.01 |
| -5.0 V | -4.99 |

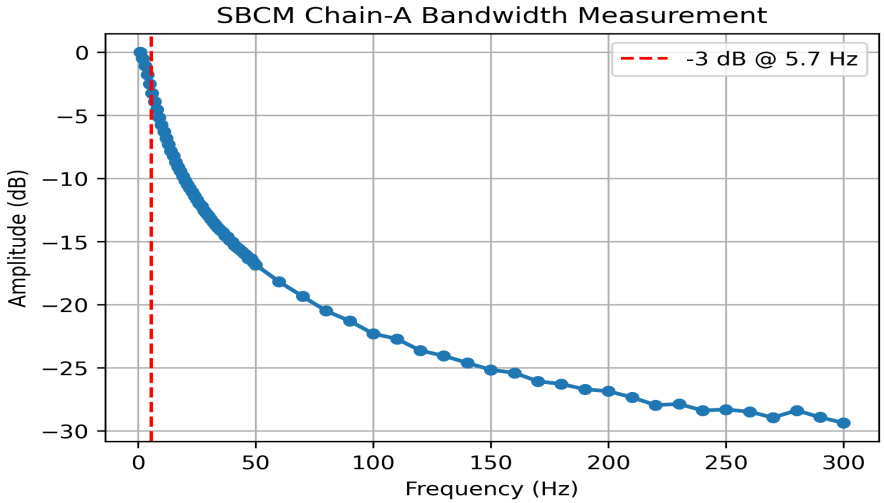
| SBCM Chain-A Bandwidth Measurement | | | | file: SBCM-A_2025_BWScan.txt | | | | |
|---|---|---|---|---|---|---|---|---|
| Freq (Hz) | Vpkpk (V) | Amp (dB) | Freq (Hz) | Vpkpk (V) | Amp (dB) | Freq (Hz) | Vpkpk (V) | Amp (dB) |
| 1 | 4.6211 | 0.0 | 26 | 1.1623 | -11.99 | 60 | 0.5689 | -18.19 |
| 2 | 4.3702 | -0.48 | 27 | 1.1344 | -12.2 | 70 | 0.498 | -19.35 |
| 3 | 4.0739 | -1.09 | 28 | 1.0881 | -12.56 | 80 | 0.4361 | -20.5 |
| 4 | 3.7605 | -1.79 | 29 | 1.0633 | -12.76 | 90 | 0.3978 | -21.3 |
| 5 | 3.4659 | -2.5 | 30 | 1.0391 | -12.96 | 100 | 0.3541 | -22.31 |
| 6 | 3.1741 | -3.26 | 31 | 1.008 | -13.23 | 110 | 0.338 | -22.72 |
| 7 | 2.9427 | -3.92 | 32 | 0.9789 | -13.48 | 120 | 0.3038 | -23.64 |
| 8 | 2.7361 | -4.55 | 33 | 0.9545 | -13.7 | 130 | 0.2894 | -24.07 |
| 9 | 2.5467 | -5.18 | 34 | 0.9311 | -13.92 | 140 | 0.2714 | -24.62 |
| 10 | 2.3802 | -5.76 | 35 | 0.9119 | -14.1 | 150 | 0.2547 | -25.17 |
| 11 | 2.2359 | -6.31 | 36 | 0.8925 | -14.28 | 160 | 0.2478 | -25.41 |
| 12 | 2.1067 | -6.82 | 37 | 0.8656 | -14.55 | 170 | 0.2294 | -26.08 |
| 13 | 1.9905 | -7.32 | 38 | 0.8573 | -14.63 | 180 | 0.2238 | -26.3 |
| 14 | 1.8789 | -7.82 | 39 | 0.8323 | -14.89 | 190 | 0.2133 | -26.72 |
| 15 | 1.7931 | -8.22 | 40 | 0.8173 | -15.05 | 200 | 0.2095 | -26.87 |
| 16 | 1.697 | -8.7 | 41 | 0.7938 | -15.3 | 210 | 0.1981 | -27.36 |
| 17 | 1.6225 | -9.09 | 42 | 0.7789 | -15.47 | 220 | 0.1847 | -27.97 |
| 18 | 1.5673 | -9.39 | 43 | 0.7677 | -15.59 | 230 | 0.1864 | -27.89 |
| 19 | 1.4909 | -9.83 | 44 | 0.752 | -15.77 | 240 | 0.1758 | -28.4 |
| 20 | 1.4338 | -10.17 | 45 | 0.7375 | -15.94 | 250 | 0.1773 | -28.32 |
| 21 | 1.3825 | -10.48 | 46 | 0.7259 | -16.08 | 260 | 0.1737 | -28.5 |
| 22 | 1.3362 | -10.78 | 47 | 0.7023 | -16.36 | 270 | 0.1647 | -28.96 |
| 23 | 1.2888 | -11.09 | 48 | 0.7022 | -16.37 | 280 | 0.1759 | -28.39 |
| 24 | 1.2433 | -11.4 | 49 | 0.6795 | -16.65 | 290 | 0.1652 | -28.94 |
| 25 | 1.2069 | -11.66 | 50 | 0.6648 | -16.84 | 300 | 0.1569 | -29.38 |

The RF input is set to 499.68 MHz @ 0 dBm.

Freq: Amplitude Modulation Hz between 1 Hz and 300 Hz.

Vpkpk: Measured voltage (pk-pk) of the demodulated signal coming from the Recitifier Module.

Amp: Measured amplitude (dB) of the demodulated signal coming from the Recitifer Module.
0 dB is referring to the first point of the data.



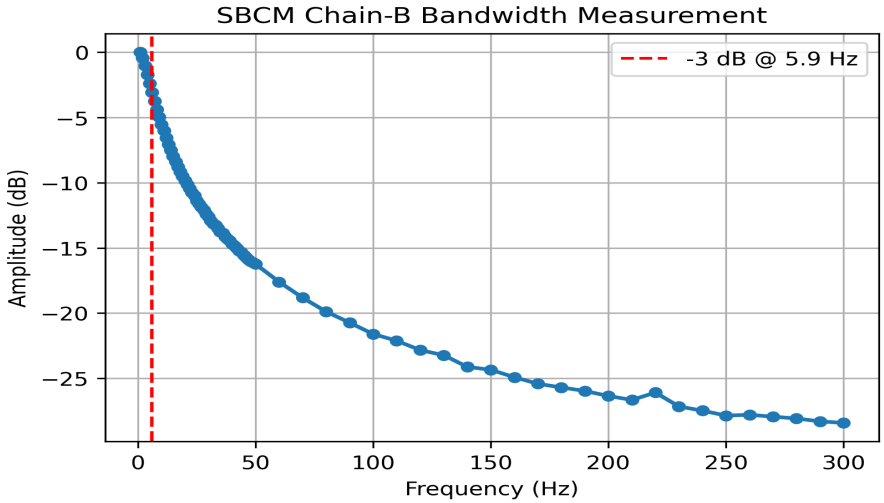
| SBCM Chain-B Bandwidth Measurement | | | | file: SBCM-B_2025_BWScan.txt | | | | |
|---|---|---|---|---|---|---|---|---|
| Freq (Hz) | Vpkpk (V) | Amp (dB) | Freq (Hz) | Vpkpk (V) | Amp (dB) | Freq (Hz) | Vpkpk (V) | Amp (dB) |
| 1 | 4.4314 | 0.0 | 26 | 1.1598 | -11.64 | 60 | 0.5839 | -17.6 |
| 2 | 4.2094 | -0.45 | 27 | 1.1281 | -11.88 | 70 | 0.5078 | -18.82 |
| 3 | 3.9292 | -1.04 | 28 | 1.1006 | -12.1 | 80 | 0.4491 | -19.88 |
| 4 | 3.6377 | -1.71 | 29 | 1.0634 | -12.4 | 90 | 0.4073 | -20.73 |
| 5 | 3.3641 | -2.39 | 30 | 1.0353 | -12.63 | 100 | 0.3684 | -21.6 |
| 6 | 3.1111 | -3.07 | 31 | 1.0014 | -12.92 | 110 | 0.3473 | -22.12 |
| 7 | 2.8777 | -3.75 | 32 | 0.9755 | -13.15 | 120 | 0.3197 | -22.84 |
| 8 | 2.6722 | -4.39 | 33 | 0.9625 | -13.26 | 130 | 0.3056 | -23.23 |
| 9 | 2.497 | -4.98 | 34 | 0.9381 | -13.49 | 140 | 0.2758 | -24.12 |
| 10 | 2.3381 | -5.55 | 35 | 0.9086 | -13.76 | 150 | 0.2688 | -24.34 |
| 11 | 2.2138 | -6.03 | 36 | 0.8983 | -13.86 | 160 | 0.2517 | -24.91 |
| 12 | 2.0822 | -6.56 | 37 | 0.8709 | -14.13 | 170 | 0.2375 | -25.42 |
| 13 | 1.9622 | -7.08 | 38 | 0.8534 | -14.31 | 180 | 0.23 | -25.7 |
| 14 | 1.868 | -7.5 | 39 | 0.8409 | -14.44 | 190 | 0.223 | -25.97 |
| 15 | 1.7683 | -7.98 | 40 | 0.8141 | -14.72 | 200 | 0.2133 | -26.35 |
| 16 | 1.6847 | -8.4 | 41 | 0.8009 | -14.86 | 210 | 0.2059 | -26.66 |
| 17 | 1.6105 | -8.79 | 42 | 0.7878 | -15.0 | 220 | 0.2198 | -26.09 |
| 18 | 1.5442 | -9.16 | 43 | 0.7677 | -15.23 | 230 | 0.1944 | -27.16 |
| 19 | 1.4825 | -9.51 | 44 | 0.7608 | -15.31 | 240 | 0.1873 | -27.48 |
| 20 | 1.4302 | -9.82 | 45 | 0.7394 | -15.55 | 250 | 0.1794 | -27.86 |
| 21 | 1.3761 | -10.16 | 46 | 0.7262 | -15.71 | 260 | 0.1806 | -27.8 |
| 22 | 1.3327 | -10.44 | 47 | 0.7119 | -15.88 | 270 | 0.1777 | -27.94 |
| 23 | 1.2869 | -10.74 | 48 | 0.6989 | -16.04 | 280 | 0.1748 | -28.08 |
| 24 | 1.2489 | -11.0 | 49 | 0.6912 | -16.14 | 290 | 0.1705 | -28.3 |
| 25 | 1.1944 | -11.39 | 50 | 0.6833 | -16.24 | 300 | 0.1681 | -28.42 |

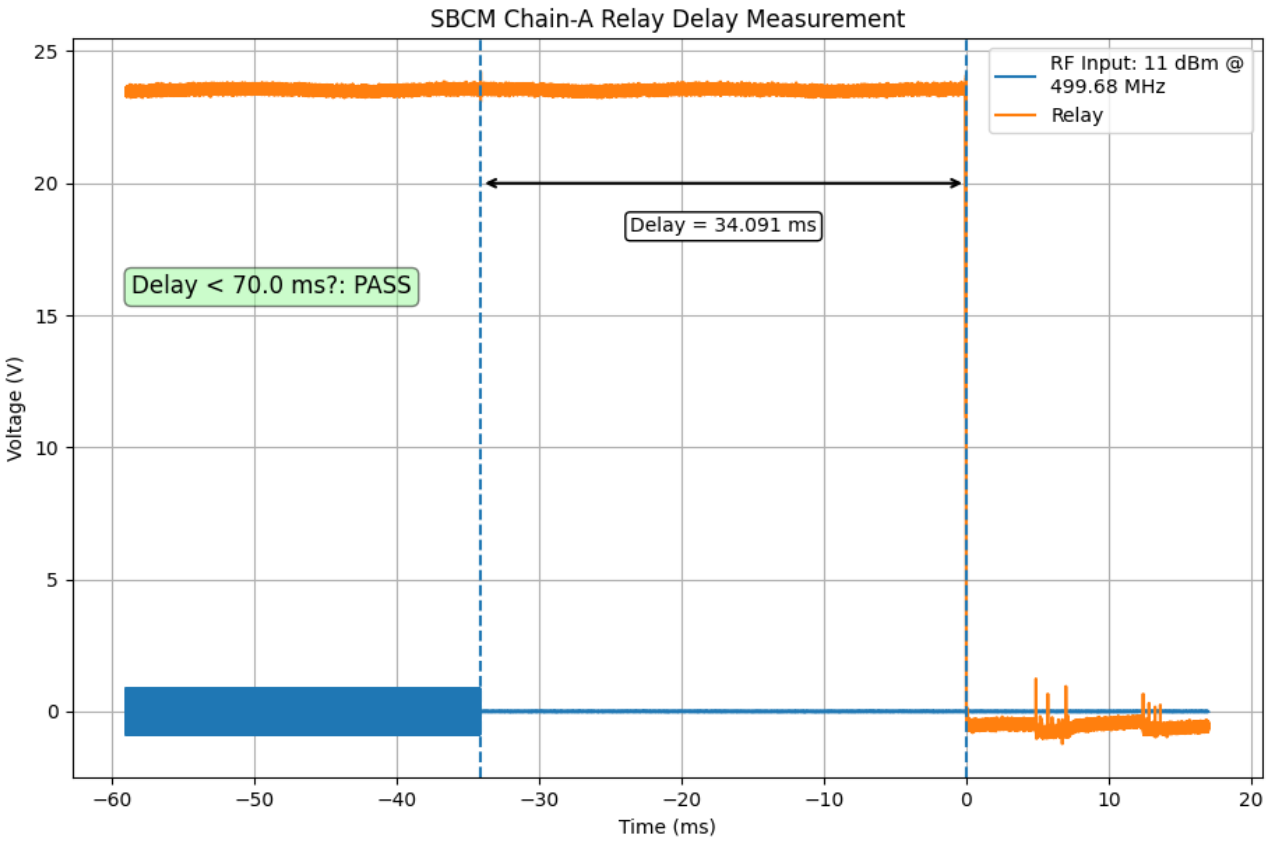
The RF input is set to 499.68 MHz @ 0 dBm.

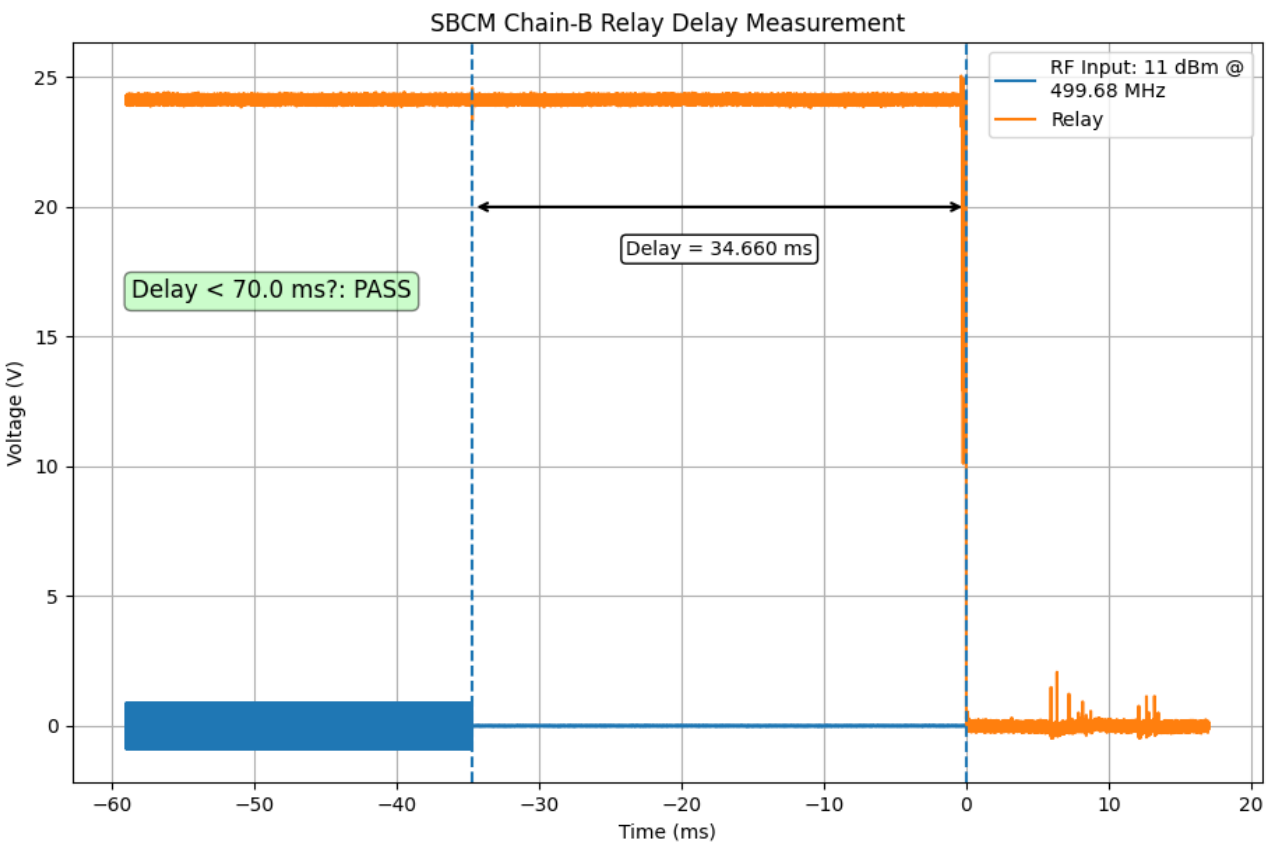
Freq: Amplitude Modulation Hz between 1 Hz and 300 Hz.

Vpkpk: Measured voltage (pk-pk) of the demodulated signal coming from the Recitifier Module.

Amp: Measured amplitude (dB) of the demodulated signal coming from the Recitifer Module.
0 dB is referring to the first point of the data.




SBCM Chain-A Relay Delay Measurement
RF Input: 11 dBm @ 499.68 MHz
Relay
Delay = 34.091 ms
Delay < 70.0 ms?: PASS
Voltage (V)
Time (ms)

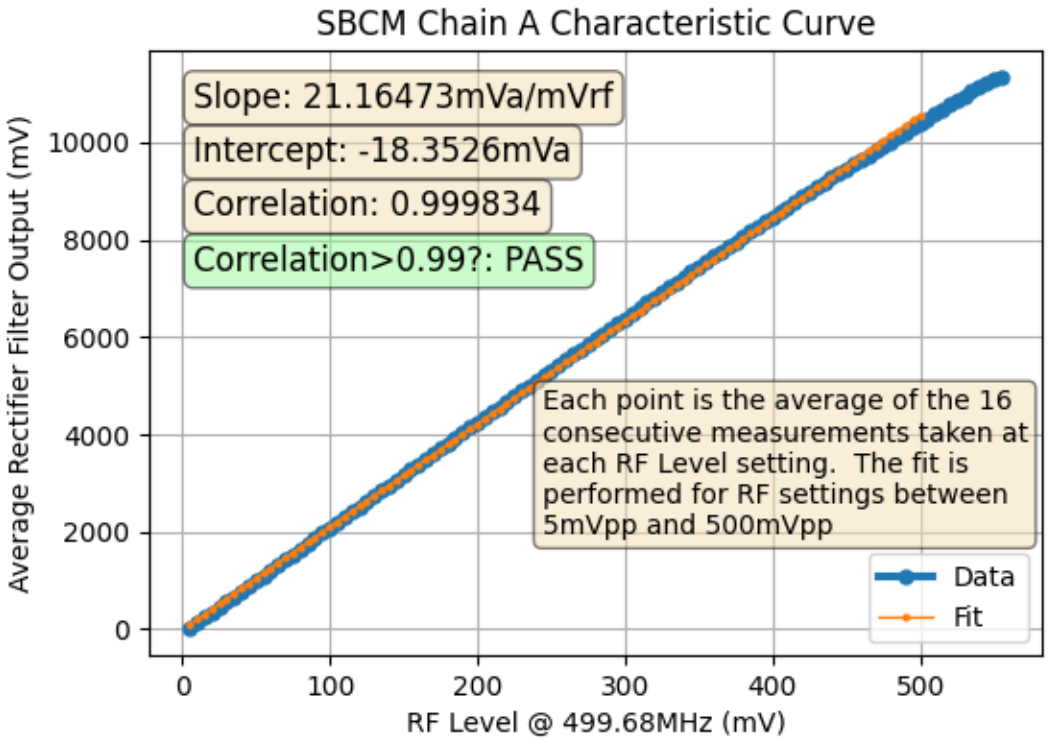

SBCM Chain-B Relay Delay Measurement
RF Input: 11 dBm @ 499.68 MHz
Relay
Delay = 34.660 ms
Delay < 70.0 ms?: PASS
Voltage (V)
Time (ms)

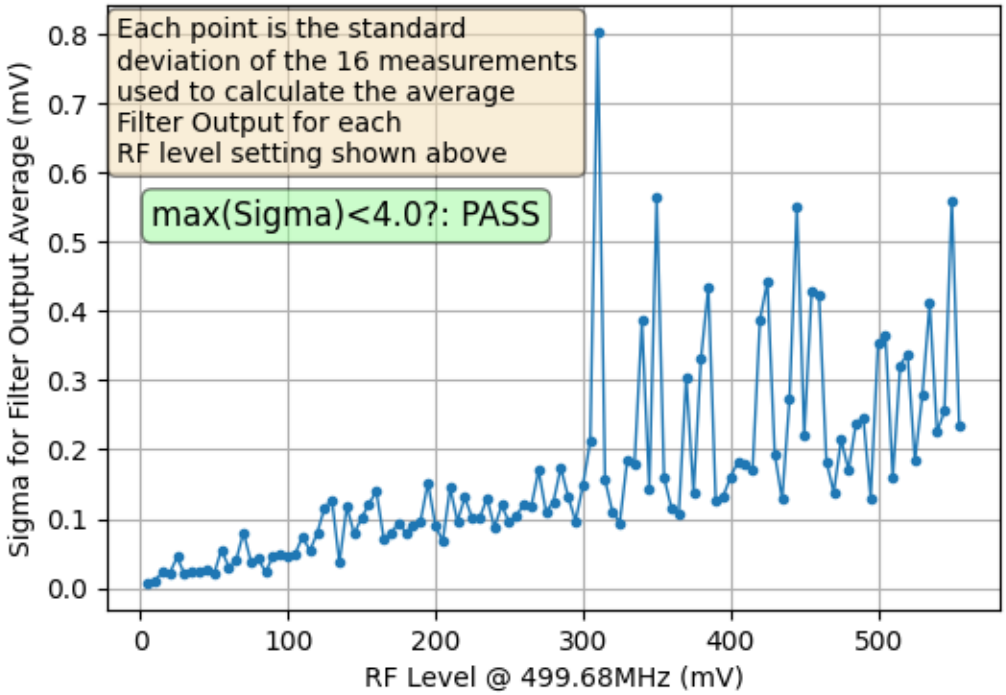
| SBCM Chain-A Characteristic Curve | | | | file: SBCM-A_2025_CharCurve.txt | | | | |
|---|---|---|---|---|---|---|---|---|
| SetPnt (mVpp) | Filter (mV) | Sigma (mV) | SetPnt (mVpp) | Filter (mV) | Sigma (mV) | SetPnt (mVpp) | Filter (mV) | Sigma (mV) |
| 5 | 27.81 | 0.0081 | 190 | 4032.07 | 0.0972 | 375 | 7960.67 | 0.137 |
| 10 | 127.62 | 0.0108 | 195 | 4139.9 | 0.1521 | 380 | 8064.41 | 0.3314 |
| 15 | 232.86 | 0.0231 | 200 | 4247.64 | 0.0912 | 385 | 8167.4 | 0.4341 |
| 20 | 339.6 | 0.0215 | 205 | 4355.28 | 0.0671 | 390 | 8271.35 | 0.1276 |
| 25 | 446.51 | 0.0447 | 210 | 4463.19 | 0.1449 | 395 | 8373.38 | 0.1323 |
| 30 | 553.8 | 0.02 | 215 | 4570.61 | 0.0965 | 400 | 8476.52 | 0.1602 |
| 35 | 662.45 | 0.0236 | 220 | 4678.14 | 0.1314 | 405 | 8577.17 | 0.1823 |
| 40 | 770.2 | 0.0239 | 225 | 4785.79 | 0.1003 | 410 | 8677.29 | 0.18 |
| 45 | 878.05 | 0.0261 | 230 | 4892.67 | 0.1009 | 415 | 8780.42 | 0.1714 |
| 50 | 986.2 | 0.0199 | 235 | 5000.11 | 0.1299 | 420 | 8882.71 | 0.3881 |
| 55 | 1094.36 | 0.0555 | 240 | 5107.07 | 0.0876 | 425 | 8984.98 | 0.4425 |
| 60 | 1204.0 | 0.0306 | 245 | 5213.61 | 0.1196 | 430 | 9085.29 | 0.1935 |
| 65 | 1312.51 | 0.0401 | 250 | 5320.6 | 0.0947 | 435 | 9180.41 | 0.128 |
| 70 | 1420.66 | 0.0805 | 255 | 5427.28 | 0.105 | 440 | 9269.77 | 0.2722 |
| 75 | 1528.72 | 0.0382 | 260 | 5534.13 | 0.121 | 445 | 9361.62 | 0.5511 |
| 80 | 1648.02 | 0.0427 | 265 | 5640.66 | 0.118 | 450 | 9454.25 | 0.2211 |
| 85 | 1756.15 | 0.0226 | 270 | 5747.62 | 0.1701 | 455 | 9547.15 | 0.4285 |
| 90 | 1865.06 | 0.0467 | 275 | 5853.73 | 0.1102 | 460 | 9640.22 | 0.4229 |
| 95 | 1973.48 | 0.0494 | 280 | 5959.85 | 0.1245 | 465 | 9733.2 | 0.1827 |
| 100 | 2081.95 | 0.0464 | 285 | 6066.17 | 0.1742 | 470 | 9827.16 | 0.1383 |
| 105 | 2191.08 | 0.0491 | 290 | 6172.19 | 0.1322 | 475 | 9922.57 | 0.2145 |
| 110 | 2299.1 | 0.0745 | 295 | 6277.98 | 0.0967 | 480 | 10016.89 | 0.1706 |
| 115 | 2408.03 | 0.0538 | 300 | 6383.38 | 0.1484 | 485 | 10110.56 | 0.237 |
| 120 | 2516.82 | 0.0799 | 305 | 6489.21 | 0.2127 | 490 | 10205.42 | 0.2471 |
| 125 | 2625.46 | 0.1166 | 310 | 6605.06 | 0.8024 | 495 | 10300.69 | 0.1303 |
| 130 | 2734.24 | 0.1264 | 315 | 6709.52 | 0.1563 | 500 | 10395.04 | 0.3531 |
| 135 | 2842.71 | 0.0364 | 320 | 6814.08 | 0.11 | 505 | 10488.41 | 0.3663 |
| 140 | 2951.32 | 0.1177 | 325 | 6919.62 | 0.0945 | 510 | 10582.86 | 0.16 |
| 145 | 3059.22 | 0.0784 | 330 | 7023.12 | 0.1846 | 515 | 10677.31 | 0.3212 |
| 150 | 3167.74 | 0.1015 | 335 | 7127.41 | 0.1801 | 520 | 10772.24 | 0.3386 |
| 155 | 3275.86 | 0.1209 | 340 | 7237.97 | 0.386 | 525 | 10866.15 | 0.1837 |
| 160 | 3384.6 | 0.1402 | 345 | 7342.11 | 0.1442 | 530 | 10961.22 | 0.279 |
| 165 | 3492.45 | 0.072 | 350 | 7445.34 | 0.5637 | 535 | 11054.48 | 0.4118 |
| 170 | 3600.57 | 0.0791 | 355 | 7549.55 | 0.1595 | 540 | 11145.06 | 0.2253 |
| 175 | 3708.68 | 0.094 | 360 | 7653.38 | 0.1156 | 545 | 11227.95 | 0.256 |
| 180 | 3816.7 | 0.0803 | 365 | 7755.3 | 0.1076 | 550 | 11282.44 | 0.5604 |
| 185 | 3924.44 | 0.0906 | 370 | 7857.35 | 0.3053 | 555 | 11328.58 | 0.2333 |

SetPnt: Rf Source Level Setting in mVpp

Filter: Average of 16 DMM measurements of the Filter Output from the Rectifier Module

Sigma: The Standard Deviation of the 16 DMM Filter measurements


SBCM Chain A Characteristic Curve
Slope: 21.16473mVa/mVrf
Intercept: -18.3526mVa
Correlation: 0.999834
Correlation>0.99?: PASS
Each point is the average of the 16 consecutive measurements taken at each RF Level setting. The fit is performed for RF settings between 5mVpp and 500mVpp
Data
Fit
Average Rectifier Filter Output (mV)
RF Level @ 499.68MHz (mV)
0
2000
4000
6000
8000
10000
0
100
200
300
400
500

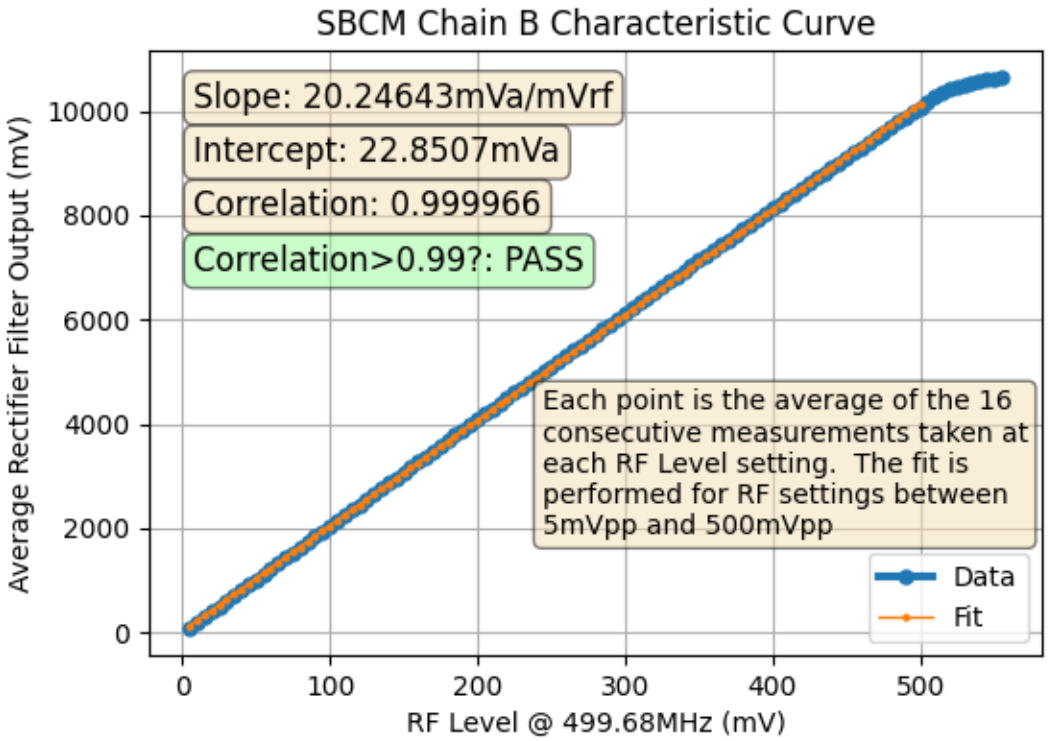

Each point is the standard deviation of the 16 measurements used to calculate the average Filter Output for each RF level setting shown above
max(Sigma)<4.0?: PASS
Sigma for Filter Output Average (mV)
RF Level @ 499.68MHz (mV)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
100
200
300
400
500

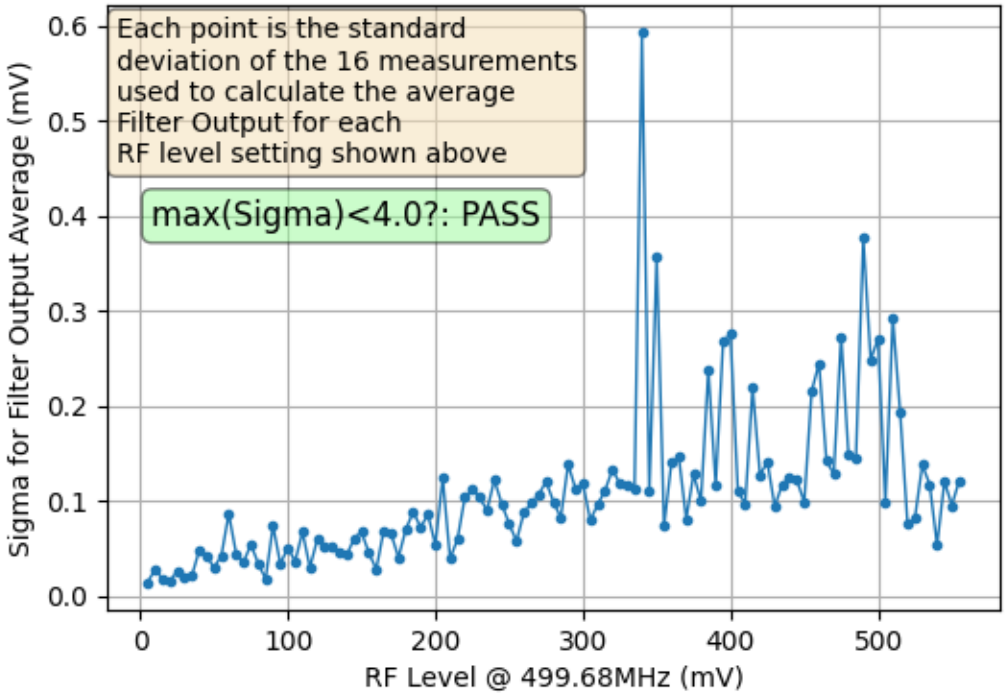
| SBCM Chain-B Characteristic Curve | | | | file: SBCM-B_2025_CharCurve.txt | | | | |
|---|---|---|---|---|---|---|---|---|
| SetPnt (mV) | Filter (mV) | Sigma (mV) | SetPnt (mV) | Filter (mV) | Sigma (mV) | SetPnt (mV) | Filter (mV) | Sigma (mV) |
| 5 | 90.33 | 0.0142 | 190 | 3884.62 | 0.0722 | 375 | 7631.88 | 0.128 |
| 10 | 190.55 | 0.0279 | 195 | 3987.05 | 0.0855 | 380 | 7730.74 | 0.0999 |
| 15 | 292.2 | 0.0184 | 200 | 4088.82 | 0.054 | 385 | 7830.48 | 0.2375 |
| 20 | 394.45 | 0.0154 | 205 | 4190.85 | 0.1241 | 390 | 7930.4 | 0.1174 |
| 25 | 496.62 | 0.0259 | 210 | 4292.82 | 0.0391 | 395 | 8028.89 | 0.2677 |
| 30 | 599.02 | 0.0201 | 215 | 4394.82 | 0.0597 | 400 | 8128.87 | 0.2756 |
| 35 | 702.67 | 0.0212 | 220 | 4496.58 | 0.1038 | 405 | 8226.58 | 0.1101 |
| 40 | 805.37 | 0.0481 | 225 | 4598.43 | 0.112 | 410 | 8323.39 | 0.097 |
| 45 | 908.16 | 0.042 | 230 | 4700.25 | 0.1037 | 415 | 8422.7 | 0.2189 |
| 50 | 1011.07 | 0.0298 | 235 | 4802.02 | 0.0901 | 420 | 8521.27 | 0.1271 |
| 55 | 1113.89 | 0.0422 | 240 | 4903.78 | 0.1221 | 425 | 8618.59 | 0.1412 |
| 60 | 1218.13 | 0.0865 | 245 | 5005.12 | 0.0969 | 430 | 8718.55 | 0.094 |
| 65 | 1321.23 | 0.0448 | 250 | 5106.72 | 0.0758 | 435 | 8817.06 | 0.1172 |
| 70 | 1424.1 | 0.0353 | 255 | 5208.17 | 0.0577 | 440 | 8912.37 | 0.1251 |
| 75 | 1526.82 | 0.0543 | 260 | 5309.78 | 0.0876 | 445 | 9010.32 | 0.1217 |
| 80 | 1631.2 | 0.0341 | 265 | 5411.14 | 0.0976 | 450 | 9109.02 | 0.0986 |
| 85 | 1733.45 | 0.0179 | 270 | 5512.43 | 0.1063 | 455 | 9206.22 | 0.2153 |
| 90 | 1836.32 | 0.074 | 275 | 5613.5 | 0.1205 | 460 | 9304.38 | 0.2447 |
| 95 | 1938.78 | 0.0335 | 280 | 5714.57 | 0.0993 | 465 | 9400.91 | 0.1427 |
| 100 | 2041.0 | 0.0507 | 285 | 5815.8 | 0.0822 | 470 | 9498.53 | 0.128 |
| 105 | 2144.12 | 0.0365 | 290 | 5916.78 | 0.1385 | 475 | 9596.49 | 0.2729 |
| 110 | 2246.16 | 0.0674 | 295 | 6017.71 | 0.1126 | 480 | 9693.81 | 0.148 |
| 115 | 2349.17 | 0.03 | 300 | 6118.61 | 0.1182 | 485 | 9789.81 | 0.1443 |
| 120 | 2451.76 | 0.0594 | 305 | 6219.61 | 0.0796 | 490 | 9887.08 | 0.3764 |
| 125 | 2553.99 | 0.0525 | 310 | 6329.8 | 0.0961 | 495 | 9984.46 | 0.2484 |
| 130 | 2656.58 | 0.0529 | 315 | 6429.88 | 0.1109 | 500 | 10080.8 | 0.2707 |
| 135 | 2759.16 | 0.0449 | 320 | 6530.16 | 0.1331 | 505 | 10177.02 | 0.0984 |
| 140 | 2861.67 | 0.0429 | 325 | 6631.09 | 0.1183 | 510 | 10270.75 | 0.2925 |
| 145 | 2963.83 | 0.0609 | 330 | 6730.19 | 0.1168 | 515 | 10352.73 | 0.1928 |
| 150 | 3066.51 | 0.0671 | 335 | 6830.3 | 0.1135 | 520 | 10416.52 | 0.0755 |
| 155 | 3168.64 | 0.0451 | 340 | 6937.82 | 0.5927 | 525 | 10466.18 | 0.0822 |
| 160 | 3271.5 | 0.0281 | 345 | 7038.19 | 0.1101 | 530 | 10507.21 | 0.1378 |
| 165 | 3373.83 | 0.0688 | 350 | 7135.94 | 0.3561 | 535 | 10542.36 | 0.1173 |
| 170 | 3476.06 | 0.0654 | 355 | 7237.0 | 0.0739 | 540 | 10572.7 | 0.0536 |
| 175 | 3578.21 | 0.0405 | 360 | 7336.81 | 0.1407 | 545 | 10599.7 | 0.1206 |
| 180 | 3680.65 | 0.0699 | 365 | 7434.83 | 0.1461 | 550 | 10622.33 | 0.0953 |
| 185 | 3782.74 | 0.0886 | 370 | 7531.92 | 0.0812 | 555 | 10640.44 | 0.1213 |

SetPnt: Rf Source Level Setting in mVpp

Filter: Average of 16 DMM measurements of the Filter Output from the Rectifier Module

Sigma: The Standard Deviation of the 16 DMM Filter measurements


SBCM Chain B Characteristic Curve
Slope: 20.24643mVa/mVrf
Intercept: 22.8507mVa
Correlation: 0.999966
Correlation>0.99?: PASS
Each point is the average of the 16 consecutive measurements taken at each RF Level setting. The fit is performed for RF settings between 5mVpp and 500mVpp
Data
Fit
Average Rectifier Filter Output (mV)
RF Level @ 499.68MHz (mV)
0
2000
4000
6000
8000
10000
0
100
200
300
400
500


Each point is the standard deviation of the 16 measurements used to calculate the average Filter Output for each RF level setting shown above
max(Sigma)<4.0?: PASS
Sigma for Filter Output Average (mV)
RF Level @ 499.68MHz (mV)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0
100
200
300
400
500

SBCM Beam Scan with 90% Fill

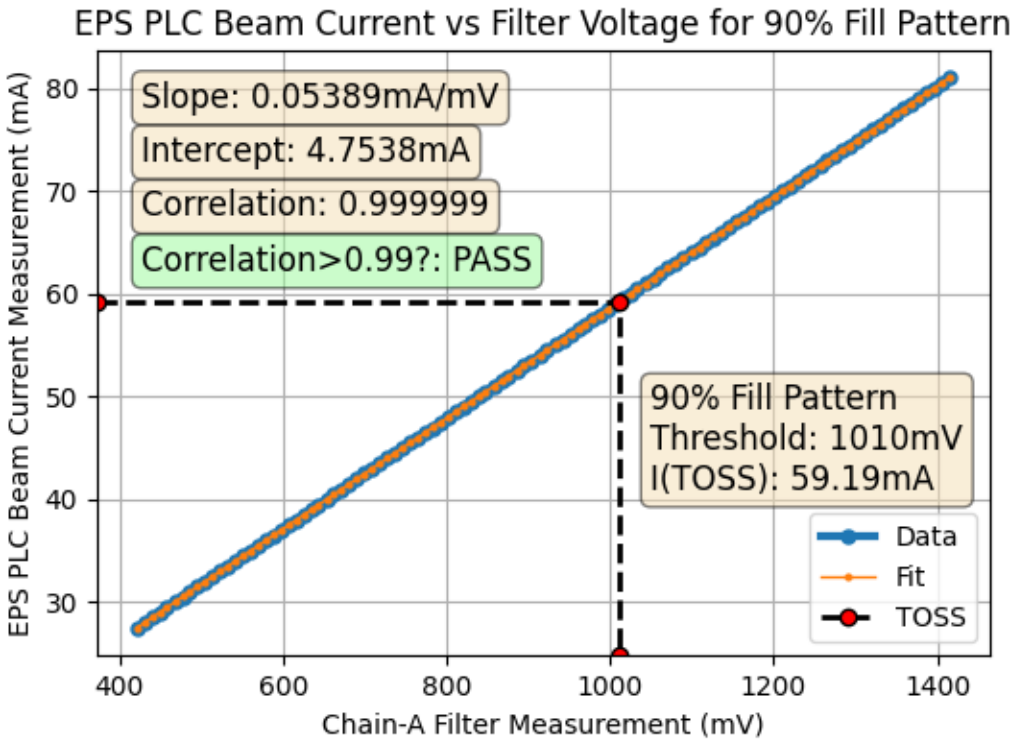
file: SBCM_BeamScanFull_90pctFill_2025.txt

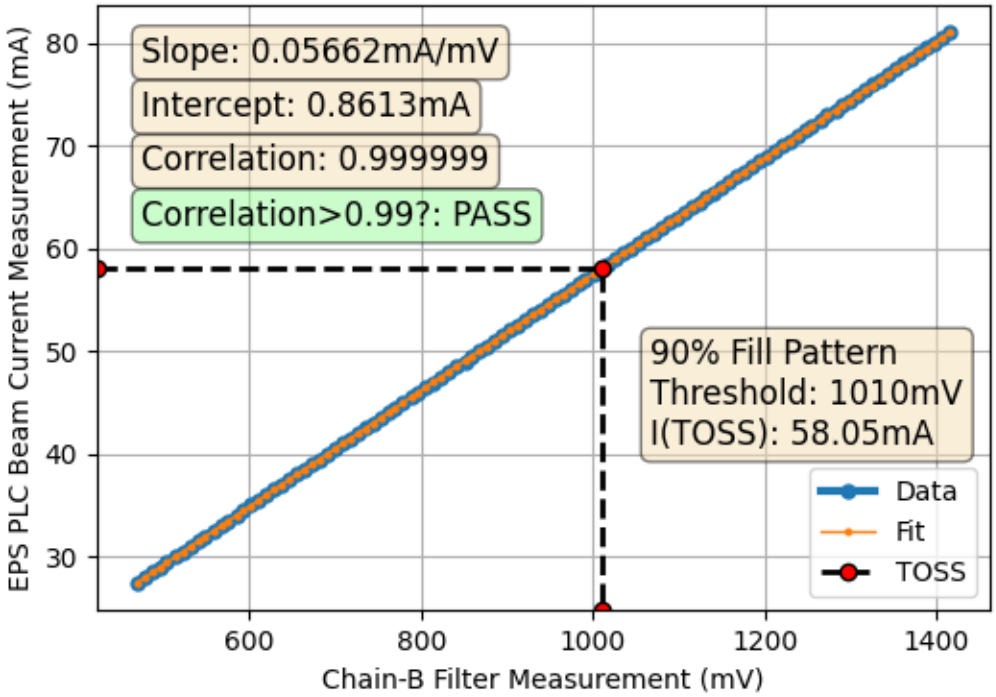
| SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) | SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) | SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) |
|---|---|---|---|---|---|---|---|---|
| 81.0 | 1414.4 | 1414.69 | 62.95 | 1079.71 | 1096.63 | 44.97 | 745.79 | 779.15 |
| 80.43 | 1404.51 | 1404.99 | 62.48 | 1070.89 | 1088.35 | 44.44 | 736.54 | 770.16 |
| 79.96 | 1395.22 | 1396.21 | 61.98 | 1061.27 | 1079.46 | 43.97 | 727.76 | 761.71 |
| 79.5 | 1386.63 | 1388.07 | 61.48 | 1052.95 | 1071.31 | 43.47 | 718.31 | 752.56 |
| 78.96 | 1376.65 | 1378.58 | 60.98 | 1042.97 | 1061.97 | 42.97 | 709.23 | 743.95 |
| 78.5 | 1368.09 | 1370.41 | 60.45 | 1033.22 | 1052.71 | 42.47 | 699.1 | 734.53 |
| 78.0 | 1358.27 | 1361.27 | 59.99 | 1024.37 | 1044.16 | 41.94 | 690.05 | 725.95 |
| 77.5 | 1349.8 | 1353.27 | 59.49 | 1015.04 | 1035.38 | 41.48 | 681.01 | 717.35 |
| 77.0 | 1340.5 | 1344.42 | 58.99 | 1006.06 | 1026.82 | 40.98 | 671.6 | 708.54 |
| 76.47 | 1330.78 | 1335.37 | 58.49 | 997.0 | 1017.87 | 40.48 | 662.67 | 700.02 |
| 75.93 | 1320.14 | 1325.46 | 57.96 | 987.53 | 1008.64 | 39.98 | 653.5 | 690.96 |
| 75.4 | 1310.71 | 1316.54 | 57.49 | 978.93 | 1000.56 | 39.44 | 643.42 | 681.46 |
| 74.94 | 1302.09 | 1308.31 | 56.99 | 969.6 | 991.41 | 38.98 | 635.26 | 673.67 |
| 74.4 | 1292.46 | 1299.13 | 56.49 | 960.99 | 982.97 | 38.48 | 625.21 | 664.2 |
| 73.94 | 1283.92 | 1290.93 | 55.99 | 951.37 | 973.84 | 37.98 | 616.61 | 655.88 |
| 73.47 | 1274.76 | 1282.14 | 55.46 | 941.99 | 964.93 | 37.48 | 607.17 | 646.84 |
| 72.94 | 1264.75 | 1272.85 | 54.99 | 932.74 | 956.39 | 36.95 | 597.64 | 637.45 |
| 72.47 | 1256.49 | 1264.98 | 54.49 | 922.91 | 947.16 | 36.48 | 588.93 | 629.23 |
| 71.91 | 1245.97 | 1255.02 | 53.99 | 914.41 | 939.22 | 35.98 | 579.3 | 619.87 |
| 71.47 | 1237.64 | 1247.26 | 53.49 | 904.13 | 929.43 | 35.48 | 570.04 | 611.2 |
| 70.97 | 1228.84 | 1238.57 | 52.96 | 894.68 | 920.65 | 34.98 | 560.67 | 602.35 |
| 70.44 | 1219.25 | 1229.33 | 52.49 | 885.85 | 912.51 | 34.45 | 551.07 | 592.94 |
| 69.94 | 1209.83 | 1220.28 | 52.0 | 876.07 | 903.05 | 33.98 | 542.7 | 585.23 |
| 69.47 | 1200.19 | 1211.23 | 51.5 | 867.34 | 894.96 | 33.48 | 532.7 | 575.56 |
| 68.98 | 1191.74 | 1203.14 | 51.0 | 857.58 | 885.46 | 32.98 | 523.94 | 567.11 |
| 68.48 | 1182.38 | 1194.02 | 50.47 | 847.65 | 876.25 | 32.49 | 514.38 | 558.16 |
| 67.94 | 1173.33 | 1185.07 | 50.0 | 838.85 | 867.72 | 31.92 | 504.01 | 548.34 |
| 67.48 | 1163.84 | 1176.06 | 49.5 | 829.55 | 858.98 | 31.49 | 496.02 | 540.66 |
| 66.98 | 1154.92 | 1167.71 | 49.0 | 821.28 | 850.76 | 30.99 | 486.62 | 531.62 |
| 66.44 | 1145.27 | 1158.25 | 48.5 | 811.72 | 841.54 | 30.49 | 478.08 | 523.33 |
| 65.95 | 1135.68 | 1149.32 | 47.97 | 801.9 | 832.35 | 29.99 | 467.93 | 513.58 |
| 65.45 | 1126.77 | 1140.94 | 47.47 | 792.3 | 823.13 | 29.46 | 458.25 | 504.31 |
| 64.95 | 1117.37 | 1131.99 | 46.94 | 783.08 | 814.35 | 28.99 | 449.78 | 496.12 |
| 64.48 | 1108.71 | 1123.74 | 46.47 | 773.76 | 805.73 | 28.49 | 440.35 | 487.13 |
| 63.98 | 1099.43 | 1115.2 | 45.97 | 764.24 | 796.8 | 27.99 | 431.58 | 478.67 |
| 63.48 | 1089.5 | 1105.83 | 45.47 | 755.71 | 788.25 | 27.49 | 422.48 | 470.0 |

SR Beam PLC: EPS PLC Measurement of Stored Beam Intensity

Filter ChainA: DMM measurement of Output from Chain-A Rectifier Module

Filter ChainB: DMM measurement of Output from Chain-B Rectifier Module


EPS PLC Beam Current vs Filter Voltage for 90% Fill Pattern
Slope: 0.05389mA/mV
Intercept: 4.7538mA
Correlation: 0.999999
Correlation>0.99?: PASS
90% Fill Pattern
Threshold: 1010mV
I(TOSS): 59.19mA
Data
Fit
TOSS
EPS PLC Beam Current Measurement (mA)
Chain-A Filter Measurement (mV)
80
70
60
50
40
30
400
600
800
1000
1200
1400


Slope: 0.05662mA/mV
Intercept: 0.8613mA
Correlation: 0.999999
Correlation>0.99?: PASS
90% Fill Pattern
Threshold: 1010mV
I(TOSS): 58.05mA
Data
Fit
TOSS
EPS PLC Beam Current Measurement (mA)
Chain-B Filter Measurement (mV)
80
70
60
50
40
30
600
800
1000
1200
1400

SBCM Beam Scan with 20% Fill

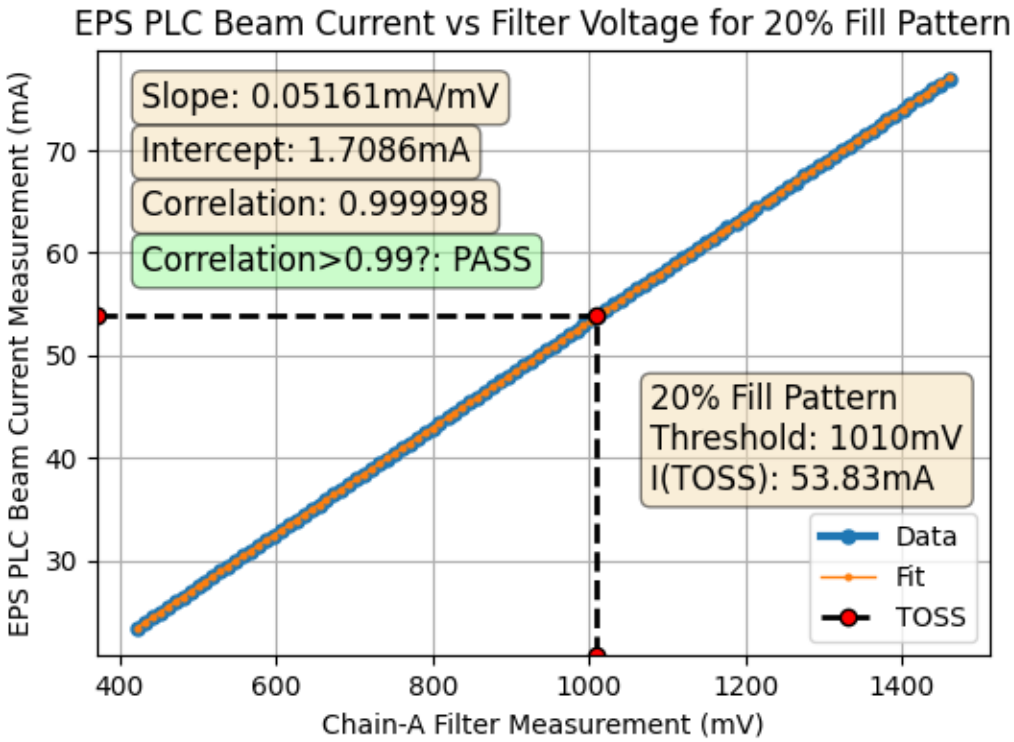
file: SBCM_BeamScanFull_20pctFill_2025.txt

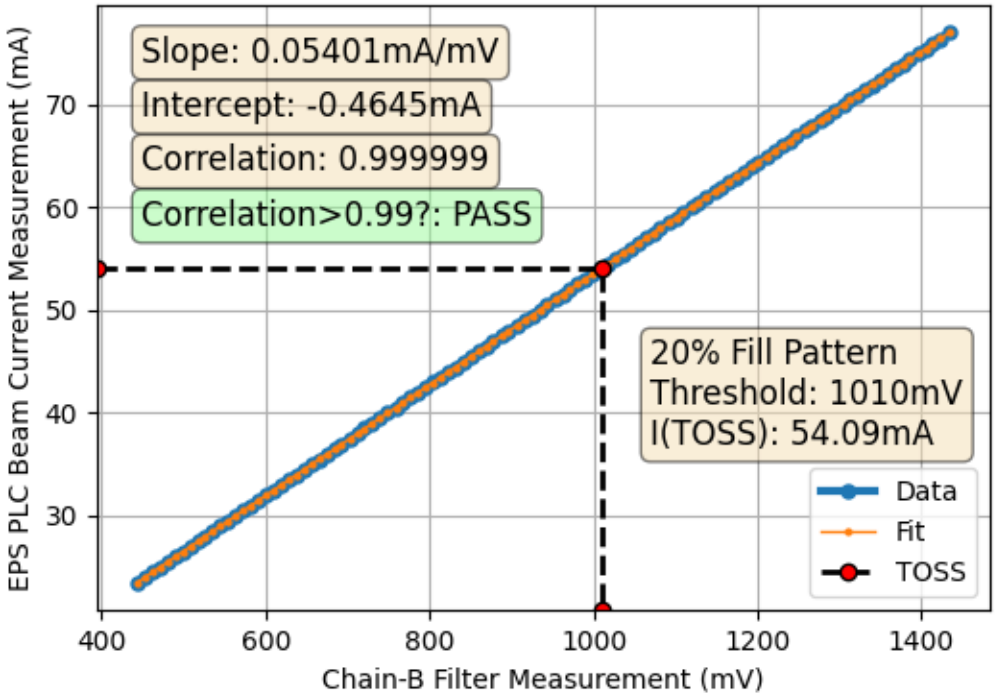
| SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) | SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) | SR Beam PLC(mA) | Chain-A Filter(mV) | Chain-B Filter(mV) |
|---|---|---|---|---|---|---|---|---|
| 77.0 | 1459.88 | 1433.68 | 58.99 | 1109.58 | 1101.26 | 40.98 | 760.56 | 766.94 |
| 76.4 | 1448.74 | 1423.31 | 58.49 | 1100.03 | 1091.63 | 40.48 | 751.74 | 758.56 |
| 75.93 | 1438.67 | 1413.75 | 57.92 | 1089.28 | 1081.23 | 39.98 | 741.85 | 748.97 |
| 75.44 | 1429.99 | 1405.31 | 57.49 | 1080.52 | 1072.94 | 39.44 | 731.4 | 738.9 |
| 74.97 | 1421.02 | 1397.23 | 56.99 | 1070.04 | 1063.46 | 38.98 | 722.11 | 730.05 |
| 74.4 | 1409.85 | 1385.95 | 56.49 | 1061.07 | 1054.69 | 38.48 | 712.15 | 720.79 |
| 73.97 | 1400.9 | 1378.15 | 55.96 | 1050.72 | 1044.93 | 37.98 | 702.75 | 712.01 |
| 73.44 | 1390.14 | 1367.87 | 55.46 | 1041.16 | 1035.82 | 37.48 | 693.02 | 702.71 |
| 72.9 | 1380.28 | 1358.51 | 54.99 | 1031.54 | 1026.68 | 36.95 | 682.68 | 692.5 |
| 72.47 | 1371.92 | 1350.1 | 54.49 | 1021.91 | 1017.59 | 36.48 | 673.47 | 683.54 |
| 71.91 | 1360.91 | 1340.2 | 53.99 | 1012.3 | 1008.48 | 35.98 | 663.37 | 674.21 |
| 71.44 | 1351.37 | 1331.1 | 53.49 | 1001.96 | 998.57 | 35.48 | 654.64 | 665.79 |
| 70.97 | 1342.6 | 1322.58 | 52.93 | 992.48 | 988.79 | 34.98 | 644.52 | 656.04 |
| 70.44 | 1331.85 | 1312.72 | 52.49 | 983.45 | 980.59 | 34.45 | 634.92 | 646.97 |
| 69.97 | 1323.16 | 1304.57 | 52.0 | 973.81 | 971.07 | 33.98 | 625.57 | 637.92 |
| 69.47 | 1312.52 | 1294.11 | 51.5 | 964.51 | 962.3 | 33.48 | 615.86 | 628.47 |
| 68.91 | 1302.08 | 1284.31 | 51.0 | 954.81 | 952.99 | 32.98 | 606.04 | 619.04 |
| 68.48 | 1293.31 | 1276.03 | 50.47 | 944.8 | 943.23 | 32.49 | 596.4 | 609.62 |
| 67.94 | 1283.35 | 1266.37 | 50.0 | 935.76 | 934.72 | 31.95 | 586.73 | 600.34 |
| 67.48 | 1274.49 | 1257.75 | 49.5 | 925.85 | 924.96 | 31.49 | 577.58 | 591.65 |
| 66.91 | 1263.76 | 1247.53 | 49.0 | 915.85 | 915.41 | 30.99 | 567.49 | 582.33 |
| 66.48 | 1254.8 | 1239.61 | 48.5 | 906.44 | 906.88 | 30.49 | 558.08 | 573.13 |
| 65.98 | 1245.16 | 1230.05 | 47.97 | 896.07 | 896.58 | 29.99 | 548.34 | 563.92 |
| 65.41 | 1234.27 | 1219.87 | 47.47 | 886.66 | 887.57 | 29.46 | 538.21 | 554.0 |
| 64.98 | 1226.25 | 1212.18 | 46.94 | 876.31 | 877.62 | 28.99 | 528.65 | 545.23 |
| 64.35 | 1213.4 | 1200.0 | 46.47 | 866.91 | 868.58 | 28.49 | 518.55 | 535.43 |
| 63.91 | 1205.32 | 1192.25 | 45.97 | 857.45 | 859.83 | 27.99 | 509.21 | 526.42 |
| 63.48 | 1196.49 | 1183.8 | 45.47 | 848.01 | 850.87 | 27.49 | 500.06 | 517.51 |
| 62.95 | 1186.71 | 1174.35 | 44.97 | 838.09 | 840.84 | 26.96 | 490.07 | 508.06 |
| 62.45 | 1176.54 | 1164.81 | 44.44 | 827.87 | 831.3 | 26.49 | 480.72 | 498.91 |
| 61.98 | 1167.74 | 1156.49 | 43.97 | 818.73 | 822.56 | 25.99 | 470.97 | 489.7 |
| 61.48 | 1158.13 | 1147.18 | 43.47 | 808.69 | 813.06 | 25.49 | 461.69 | 481.08 |
| 60.98 | 1147.53 | 1136.9 | 42.97 | 799.33 | 803.88 | 25.0 | 451.67 | 471.53 |
| 60.45 | 1138.02 | 1128.32 | 42.47 | 789.69 | 794.93 | 24.5 | 441.84 | 462.07 |
| 59.99 | 1128.84 | 1119.59 | 41.94 | 779.88 | 785.41 | 24.0 | 432.62 | 453.13 |
| 59.49 | 1118.46 | 1109.61 | 41.48 | 770.65 | 776.47 | 23.5 | 422.82 | 443.79 |

SR Beam PLC: EPS PLC Measurement of Stored Beam Intensity

Filter ChainA: DMM measurement of Output from Chain-A Rectifier Module

Filter ChainB: DMM measurement of Output from Chain-B Rectifier Module




| |
|---|
| Based on the data shown above the following changes were made: |
| I(TOSS) = 54mA for a 20% Fill Pattern |
| Chain-A Threshold set to 1010mV |
| Chain-B Threshold set to 1010mV |

SBCM Functional Scan with 20% Fill and Increasing Beam

file: SBCM_BeamScanFull_20pctFill_2025.txt

| Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R | Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R |
|---|---|---|---|---|---|---|---|---|---|
| 0.19 | 8.2 | 0,0,0 | 34.62 | 0,0,0 | 31.02 | 568.84 | 1,0,0 | 585.49 | 1,0,0 |
| 0.46 | 9.04 | 1,0,0 | 38.7 | 0,0,0 | 32.05 | 589.1 | 1,0,0 | 604.62 | 1,0,0 |
| 0.52 | 10.85 | 1,0,0 | 41.1 | 1,0,0 | 33.12 | 610.19 | 1,0,0 | 628.16 | 1,0,0 |
| 1.09 | 19.84 | 1,0,0 | 53.56 | 1,0,0 | 34.08 | 628.31 | 1,0,0 | 644.05 | 1,0,0 |
| 2.02 | 31.63 | 1,0,0 | 68.52 | 1,0,0 | 35.15 | 648.5 | 1,0,0 | 662.12 | 1,0,0 |
| 3.02 | 44.35 | 1,0,0 | 80.52 | 1,0,0 | 36.08 | 667.42 | 1,0,0 | 680.35 | 1,0,0 |
| 4.05 | 61.15 | 1,0,0 | 96.06 | 1,0,0 | 37.01 | 684.97 | 1,0,0 | 702.67 | 1,0,0 |
| 5.02 | 82.62 | 1,0,0 | 119.37 | 1,0,0 | 38.05 | 704.6 | 1,0,0 | 717.44 | 1,0,0 |
| 6.05 | 100.76 | 1,0,0 | 136.68 | 1,0,0 | 39.11 | 726.07 | 1,0,0 | 738.29 | 1,0,0 |
| 7.02 | 113.98 | 1,0,0 | 146.73 | 1,0,0 | 40.04 | 744.49 | 1,0,0 | 754.98 | 1,0,0 |
| 8.08 | 137.0 | 1,0,0 | 171.63 | 1,0,0 | 41.01 | 762.06 | 1,0,0 | 771.37 | 1,0,0 |
| 9.11 | 158.12 | 1,0,0 | 194.91 | 1,0,0 | 42.04 | 782.48 | 1,0,0 | 790.49 | 1,0,0 |
| 10.01 | 171.78 | 1,0,0 | 205.93 | 1,0,0 | 43.11 | 803.9 | 1,0,0 | 811.54 | 1,0,0 |
| 11.08 | 191.19 | 1,0,0 | 222.16 | 1,0,0 | 44.07 | 821.48 | 1,0,0 | 828.2 | 1,0,0 |
| 12.01 | 211.68 | 1,0,0 | 246.76 | 1,0,0 | 45.0 | 841.79 | 1,0,0 | 847.32 | 1,0,0 |
| 13.11 | 230.65 | 1,0,0 | 266.28 | 1,0,0 | 46.04 | 860.49 | 1,0,0 | 865.38 | 1,0,0 |
| 14.01 | 244.17 | 1,0,0 | 273.13 | 1,0,0 | 47.0 | 878.6 | 1,0,0 | 882.89 | 1,0,0 |
| 15.04 | 267.07 | 1,0,0 | 298.03 | 1,0,0 | 48.03 | 898.16 | 1,0,0 | 901.79 | 1,0,0 |
| 16.11 | 288.34 | 1,0,0 | 321.2 | 1,0,0 | 49.07 | 918.67 | 1,0,0 | 921.13 | 1,0,0 |
| 17.04 | 302.53 | 1,0,0 | 331.67 | 1,0,0 | 50.13 | 939.47 | 1,0,0 | 941.87 | 1,0,0 |
| 18.1 | 321.78 | 1,0,0 | 347.39 | 1,0,0 | 51.1 | 957.87 | 1,0,0 | 958.46 | 1,0,0 |
| 19.1 | 344.32 | 1,0,0 | 374.1 | 1,0,0 | 52.06 | 976.83 | 1,0,0 | 977.18 | 1,0,0 |
| 20.03 | 360.75 | 1,0,0 | 389.91 | 1,0,0 | 53.03 | 996.0 | 1,0,0 | 996.04 | 1,0,0 |
| 21.1 | 377.81 | 1,0,0 | 401.47 | 1,0,0 | 53.53 | 1005.25 | 1,0,0 | 1005.05 | 1,2,3 |
| 22.03 | 397.39 | 1,0,0 | 421.76 | 1,0,0 | 53.53 | 1005.29 | 1,2,3 | 1005.04 | 1,2,3 |
| 23.06 | 418.85 | 1,0,0 | 446.9 | 1,0,0 | 53.76 | 1010.68 | 1,2,3 | 1010.48 | 1,2,3 |
| 24.1 | 435.31 | 1,0,0 | 458.85 | 1,0,0 | 53.76 | 1010.68 | 1,2,3 | 1010.41 | 1,2,3 |
| 25.13 | 455.5 | 1,0,0 | 476.22 | 1,0,0 | 54.06 | 1015.52 | 1,2,3 | 1014.95 | 1,2,3 |
| 26.03 | 474.7 | 1,0,0 | 499.54 | 1,0,0 | 55.09 | 1035.39 | 1,2,3 | 1033.49 | 1,2,3 |
| 27.06 | 493.31 | 1,0,0 | 516.64 | 1,0,0 | 56.02 | 1052.92 | 1,2,3 | 1050.36 | 1,2,3 |
| 28.12 | 513.13 | 1,0,0 | 531.52 | 1,0,0 | 57.02 | 1071.62 | 1,2,3 | 1067.93 | 1,2,3 |
| 29.09 | 531.87 | 1,0,0 | 551.11 | 1,0,0 | 58.26 | 1095.85 | 1,2,3 | 1090.53 | 1,2,3 |
| 30.05 | 552.25 | 1,0,0 | 573.44 | 1,0,0 | 59.19 | 1115.04 | 1,2,3 | 1108.91 | 1,2,3 |

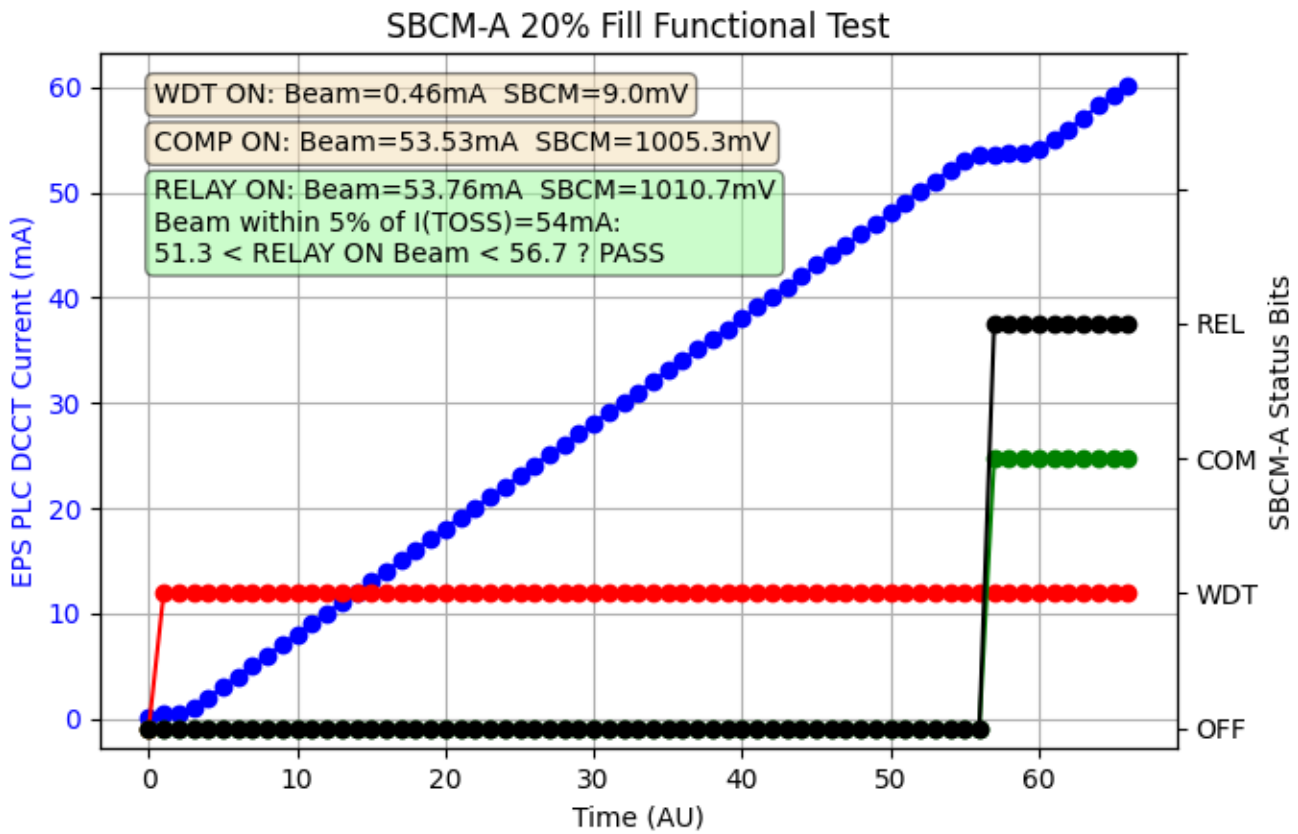
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

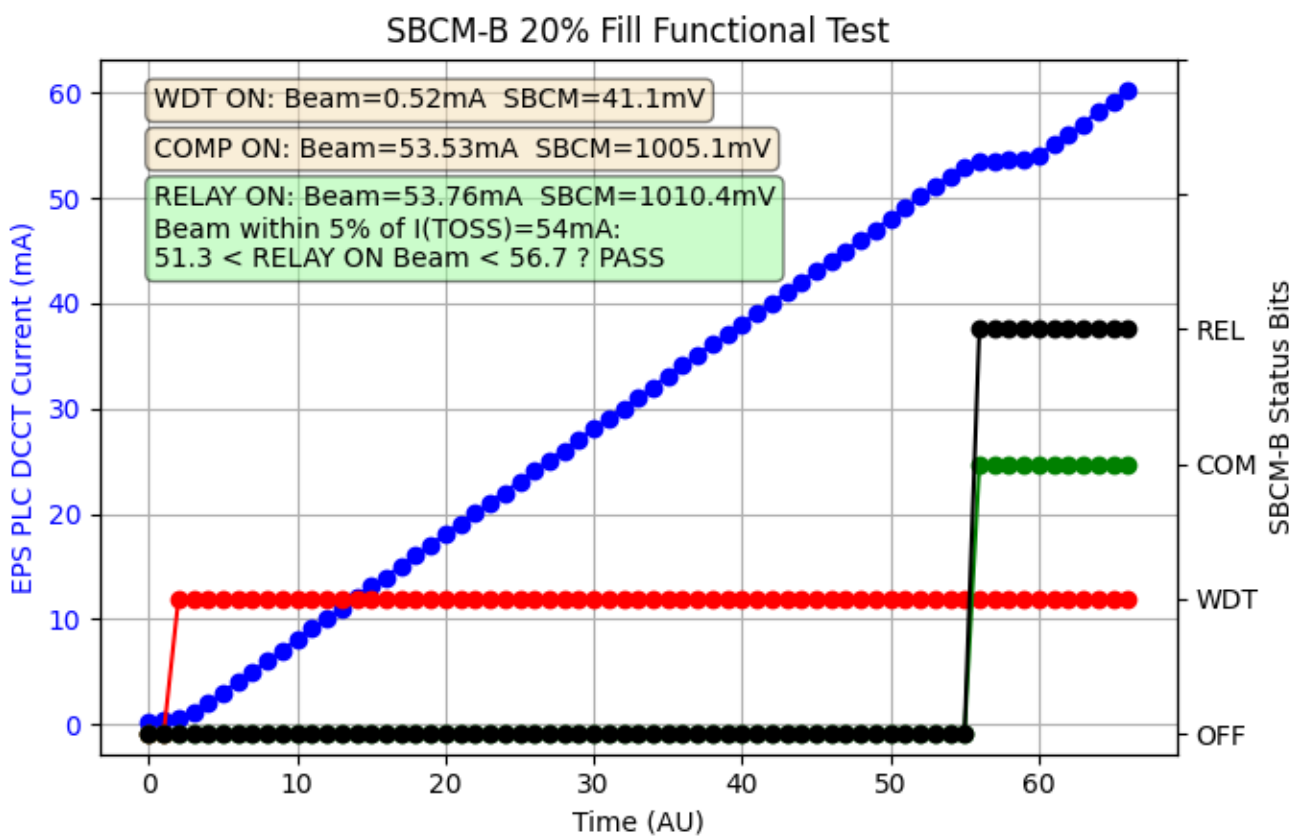
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 20% Fill Functional Test
WDT ON: Beam=0.46mA SBCM=9.0mV
COMP ON: Beam=53.53mA SBCM=1005.3mV
RELAY ON: Beam=53.76mA SBCM=1010.7mV
Beam within 5% of I(TOSS)=54mA:
51.3 < RELAY ON Beam < 56.7 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 20% Fill Functional Test
WDT ON: Beam=0.52mA SBCM=41.1mV
COMP ON: Beam=53.53mA SBCM=1005.1mV
RELAY ON: Beam=53.76mA SBCM=1010.4mV
Beam within 5% of I(TOSS)=54mA:
51.3 < RELAY ON Beam < 56.7 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 20% Fill and Decreasing Beam

file: SBCM_FunctionalTestFalling_20pctFill_2025.txt

| Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R | Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R |
|---|---|---|---|---|---|---|---|---|---|
| 59.52 | 1122.48 | 1,2,3 | 1115.31 | 1,2,3 | 30.99 | 567.79 | 1,0,0 | 583.38 | 1,0,0 |
| 58.99 | 1111.79 | 1,2,3 | 1105.19 | 1,2,3 | 29.99 | 548.97 | 1,0,0 | 565.16 | 1,0,0 |
| 57.92 | 1091.52 | 1,2,3 | 1085.49 | 1,2,3 | 28.99 | 529.9 | 1,0,0 | 546.96 | 1,0,0 |
| 56.99 | 1072.5 | 1,2,3 | 1067.41 | 1,2,3 | 27.99 | 510.91 | 1,0,0 | 528.84 | 1,0,0 |
| 55.99 | 1053.35 | 1,2,3 | 1049.11 | 1,2,3 | 26.96 | 491.05 | 1,0,0 | 509.68 | 1,0,0 |
| 54.96 | 1033.36 | 1,2,3 | 1029.93 | 1,2,3 | 25.99 | 471.53 | 1,0,0 | 491.12 | 1,0,0 |
| 53.99 | 1015.0 | 1,2,3 | 1012.56 | 1,2,3 | 25.0 | 452.49 | 1,0,0 | 472.8 | 1,0,0 |
| 53.36 | 1002.25 | 1,2,3 | 999.93 | 1,0,0 | 24.0 | 432.93 | 1,0,0 | 454.16 | 1,0,0 |
| 53.36 | 1002.25 | 1,2,3 | 999.93 | 1,0,0 | 23.0 | 414.35 | 1,0,0 | 436.2 | 1,0,0 |
| 52.93 | 994.32 | 1,2,3 | 992.68 | 1,0,0 | 22.0 | 394.22 | 1,0,0 | 417.02 | 1,0,0 |
| 52.46 | 985.29 | 1,0,0 | 983.62 | 1,0,0 | 20.97 | 375.09 | 1,0,0 | 398.61 | 1,0,0 |
| 52.46 | 985.29 | 1,0,0 | 983.62 | 1,0,0 | 19.97 | 355.94 | 1,0,0 | 380.3 | 1,0,0 |
| 52.0 | 975.71 | 1,0,0 | 974.7 | 1,0,0 | 18.97 | 337.07 | 1,0,0 | 362.07 | 1,0,0 |
| 51.0 | 956.39 | 1,0,0 | 956.07 | 1,0,0 | 17.97 | 318.71 | 1,0,0 | 344.17 | 1,0,0 |
| 50.0 | 937.0 | 1,0,0 | 937.41 | 1,0,0 | 16.94 | 298.77 | 1,0,0 | 324.98 | 1,0,0 |
| 48.97 | 917.14 | 1,0,0 | 918.56 | 1,0,0 | 15.97 | 280.23 | 1,0,0 | 306.79 | 1,0,0 |
| 47.97 | 897.74 | 1,0,0 | 899.89 | 1,0,0 | 14.97 | 261.32 | 1,0,0 | 288.79 | 1,0,0 |
| 46.94 | 877.72 | 1,0,0 | 880.58 | 1,0,0 | 13.97 | 243.08 | 1,0,0 | 270.84 | 1,0,0 |
| 45.97 | 858.87 | 1,0,0 | 862.41 | 1,0,0 | 12.98 | 224.71 | 1,0,0 | 253.31 | 1,0,0 |
| 44.97 | 839.63 | 1,0,0 | 843.89 | 1,0,0 | 11.94 | 205.38 | 1,0,0 | 234.55 | 1,0,0 |
| 43.97 | 819.89 | 1,0,0 | 825.05 | 1,0,0 | 10.98 | 187.12 | 1,0,0 | 216.77 | 1,0,0 |
| 42.97 | 800.27 | 1,0,0 | 806.25 | 1,0,0 | 9.98 | 168.46 | 1,0,0 | 198.56 | 1,0,0 |
| 41.94 | 780.78 | 1,0,0 | 787.36 | 1,0,0 | 8.98 | 150.06 | 1,0,0 | 180.97 | 1,0,0 |
| 40.98 | 761.21 | 1,0,0 | 768.85 | 1,0,0 | 7.98 | 131.91 | 1,0,0 | 163.4 | 1,0,0 |
| 39.98 | 742.09 | 1,0,0 | 750.4 | 1,0,0 | 6.92 | 111.92 | 1,0,0 | 144.12 | 1,0,0 |
| 38.98 | 723.07 | 1,0,0 | 732.17 | 1,0,0 | 5.98 | 93.98 | 1,0,0 | 126.86 | 1,0,0 |
| 37.98 | 703.91 | 1,0,0 | 713.77 | 1,0,0 | 4.99 | 76.73 | 1,0,0 | 109.83 | 1,0,0 |
| 36.95 | 683.62 | 1,0,0 | 694.14 | 1,0,0 | 3.99 | 58.65 | 1,0,0 | 92.57 | 1,0,0 |
| 35.98 | 664.74 | 1,0,0 | 676.36 | 1,0,0 | 2.99 | 40.85 | 1,0,0 | 75.11 | 1,0,0 |
| 34.98 | 645.58 | 1,0,0 | 658.08 | 1,0,0 | 1.96 | 22.66 | 1,0,0 | 57.06 | 1,0,0 |
| 33.95 | 625.9 | 1,0,0 | 639.43 | 1,0,0 | 1.12 | 9.44 | 1,0,0 | 42.92 | 0,0,0 |
| 32.98 | 606.87 | 1,0,0 | 621.07 | 1,0,0 | 1.09 | 9.24 | 0,0,0 | 42.52 | 0,0,0 |
| 31.95 | 586.8 | 1,0,0 | 601.71 | 1,0,0 | 0.99 | 8.37 | 0,0,0 | 40.64 | 0,0,0 |

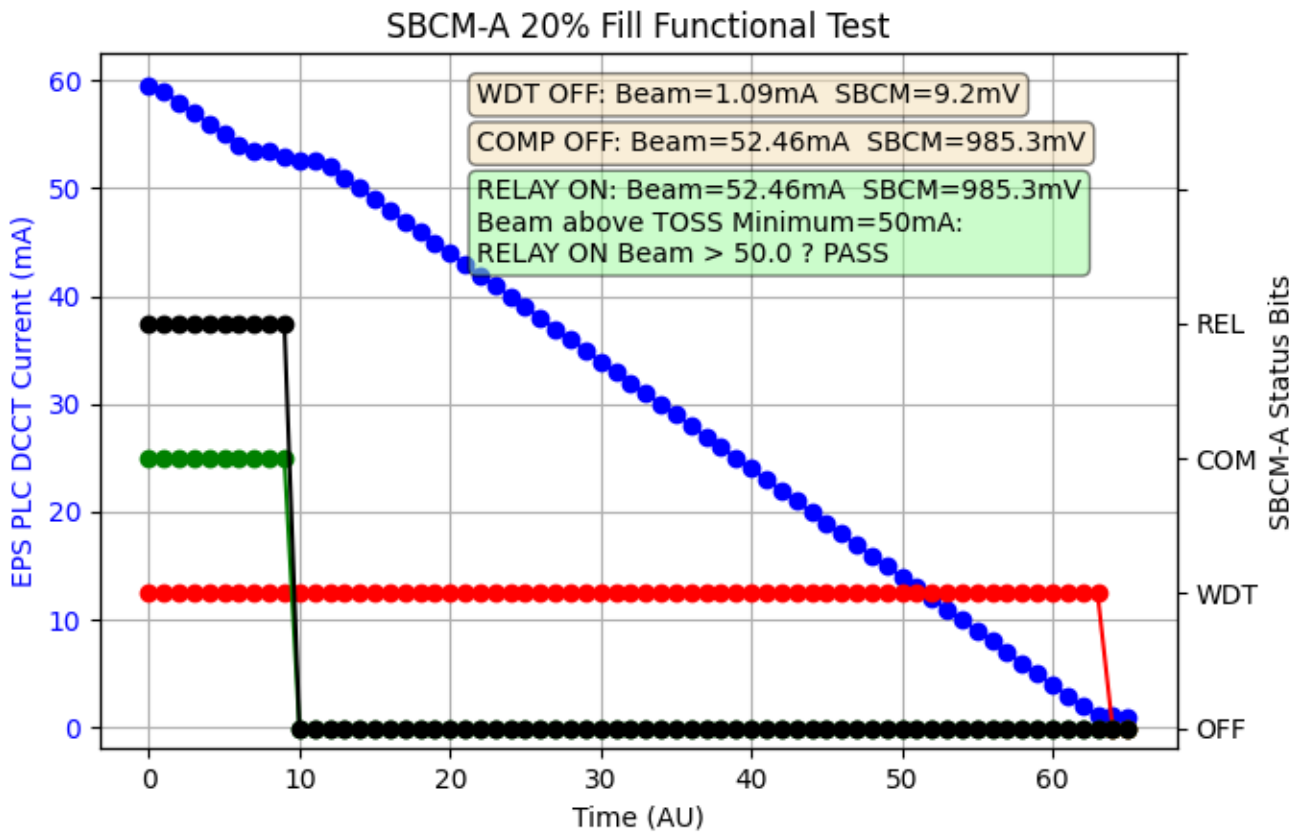
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

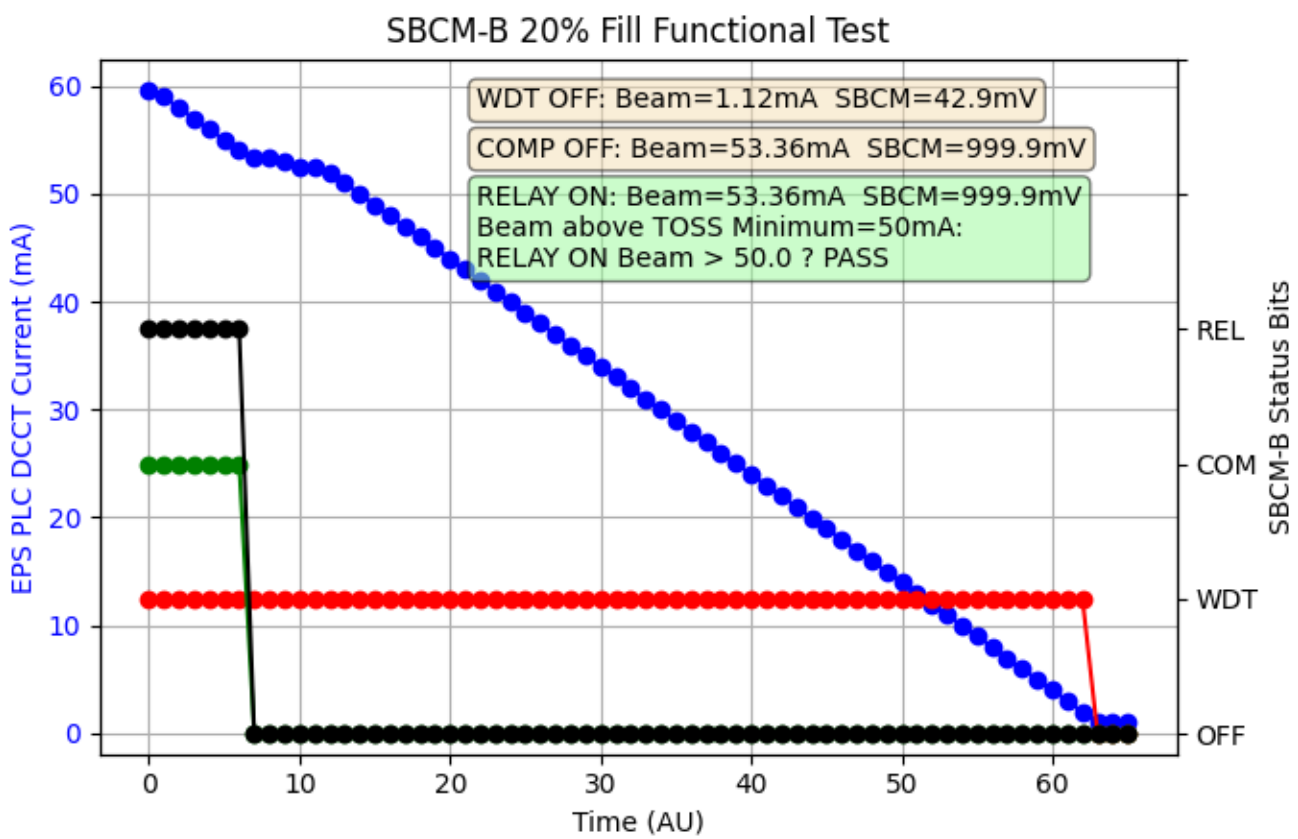
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 20% Fill Functional Test
WDT OFF: Beam=1.09mA SBCM=9.2mV
COMP OFF: Beam=52.46mA SBCM=985.3mV
RELAY ON: Beam=52.46mA SBCM=985.3mV
Beam above TOSS Minimum=50mA:
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 20% Fill Functional Test
WDT OFF: Beam=1.12mA SBCM=42.9mV
COMP OFF: Beam=53.36mA SBCM=999.9mV
RELAY ON: Beam=53.36mA SBCM=999.9mV
Beam above TOSS Minimum=50mA:
RELAY ON Beam > 50.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 90% Fill and Increasing Beam

file: SBCM_FunctionalTestRising_90pctFill_2025.txt

| Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R | Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R |
|---|---|---|---|---|---|---|---|---|---|
| 0.19 | 7.68 | 0,0,0 | 34.7 | 0,0,0 | 31.25 | 532.87 | 1,0,0 | 565.04 | 1,0,0 |
| 0.62 | 12.08 | 1,0,0 | 42.52 | 1,0,0 | 32.09 | 545.2 | 1,0,0 | 574.65 | 1,0,0 |
| 0.62 | 12.08 | 1,0,0 | 42.52 | 1,0,0 | 33.32 | 563.25 | 1,0,0 | 589.64 | 1,0,0 |
| 1.06 | 20.63 | 1,0,0 | 55.95 | 1,0,0 | 34.15 | 576.2 | 1,0,0 | 603.81 | 1,0,0 |
| 2.25 | 44.92 | 1,0,0 | 90.67 | 1,0,0 | 35.02 | 589.63 | 1,0,0 | 620.13 | 1,0,0 |
| 3.12 | 62.72 | 1,0,0 | 112.78 | 1,0,0 | 36.28 | 610.68 | 1,0,0 | 644.17 | 1,0,0 |
| 4.35 | 88.53 | 1,0,0 | 142.12 | 1,0,0 | 37.11 | 624.36 | 1,0,0 | 658.43 | 1,0,0 |
| 5.22 | 105.37 | 1,0,0 | 159.96 | 1,0,0 | 38.31 | 645.87 | 1,0,0 | 679.68 | 1,0,0 |
| 6.05 | 120.87 | 1,0,0 | 174.18 | 1,0,0 | 39.11 | 660.83 | 1,0,0 | 693.92 | 1,0,0 |
| 7.28 | 141.37 | 1,0,0 | 190.24 | 1,0,0 | 40.01 | 675.39 | 1,0,0 | 707.05 | 1,0,0 |
| 8.15 | 153.33 | 1,0,0 | 199.79 | 1,0,0 | 41.27 | 698.02 | 1,0,0 | 727.64 | 1,0,0 |
| 9.41 | 169.7 | 1,0,0 | 210.37 | 1,0,0 | 42.14 | 712.72 | 1,0,0 | 740.91 | 1,0,0 |
| 10.25 | 179.42 | 1,0,0 | 215.41 | 1,0,0 | 43.04 | 726.71 | 1,0,0 | 753.7 | 1,0,0 |
| 11.08 | 189.48 | 1,0,0 | 220.09 | 1,0,0 | 44.27 | 747.23 | 1,0,0 | 772.91 | 1,0,0 |
| 12.34 | 208.69 | 1,0,0 | 245.31 | 1,0,0 | 45.14 | 760.99 | 1,0,0 | 786.23 | 1,0,0 |
| 13.18 | 225.54 | 1,0,0 | 266.87 | 1,0,0 | 46.0 | 774.64 | 1,0,0 | 799.71 | 1,0,0 |
| 14.01 | 242.11 | 1,0,0 | 285.9 | 1,0,0 | 47.3 | 794.83 | 1,0,0 | 820.73 | 1,0,0 |
| 15.24 | 268.22 | 1,0,0 | 313.32 | 1,0,0 | 48.13 | 807.75 | 1,0,0 | 834.3 | 1,0,0 |
| 16.17 | 284.64 | 1,0,0 | 329.52 | 1,0,0 | 49.37 | 826.76 | 1,0,0 | 854.1 | 1,0,0 |
| 17.0 | 299.74 | 1,0,0 | 343.53 | 1,0,0 | 50.2 | 840.48 | 1,0,0 | 868.28 | 1,0,0 |
| 18.2 | 320.68 | 1,0,0 | 361.76 | 1,0,0 | 51.23 | 857.58 | 1,0,0 | 885.3 | 1,0,0 |
| 19.04 | 333.24 | 1,0,0 | 371.83 | 1,0,0 | 52.13 | 874.25 | 1,0,0 | 901.3 | 1,0,0 |
| 20.33 | 351.16 | 1,0,0 | 386.74 | 1,0,0 | 53.36 | 897.58 | 1,0,0 | 923.62 | 1,0,0 |
| 21.2 | 362.87 | 1,0,0 | 395.07 | 1,0,0 | 54.26 | 913.77 | 1,0,0 | 939.12 | 1,0,0 |
| 22.03 | 373.93 | 1,0,0 | 402.76 | 1,0,0 | 55.03 | 929.0 | 1,0,0 | 953.33 | 1,0,0 |
| 23.3 | 392.98 | 1,0,0 | 425.33 | 1,0,0 | 56.29 | 952.89 | 1,0,0 | 975.12 | 1,0,0 |
| 24.16 | 407.73 | 1,0,0 | 444.19 | 1,0,0 | 57.09 | 967.79 | 1,0,0 | 988.51 | 1,0,0 |
| 25.43 | 431.14 | 1,0,0 | 470.56 | 1,0,0 | 57.89 | 982.54 | 1,0,0 | 1002.08 | 1,2,3 |
| 26.29 | 447.37 | 1,0,0 | 487.38 | 1,0,0 | 57.89 | 982.54 | 1,0,0 | 1002.08 | 1,2,3 |
| 27.13 | 463.27 | 1,0,0 | 503.34 | 1,0,0 | 58.26 | 989.1 | 1,0,0 | 1008.58 | 1,2,3 |
| 28.36 | 485.5 | 1,0,0 | 523.95 | 1,0,0 | 59.05 | 1003.37 | 1,0,0 | 1022.07 | 1,2,3 |
| 29.19 | 499.56 | 1,0,0 | 536.51 | 1,0,0 | 59.42 | 1010.95 | 1,2,3 | 1029.42 | 1,2,3 |
| 30.02 | 513.28 | 1,0,0 | 548.52 | 1,0,0 | 59.42 | 1010.95 | 1,2,3 | 1029.42 | 1,2,3 |

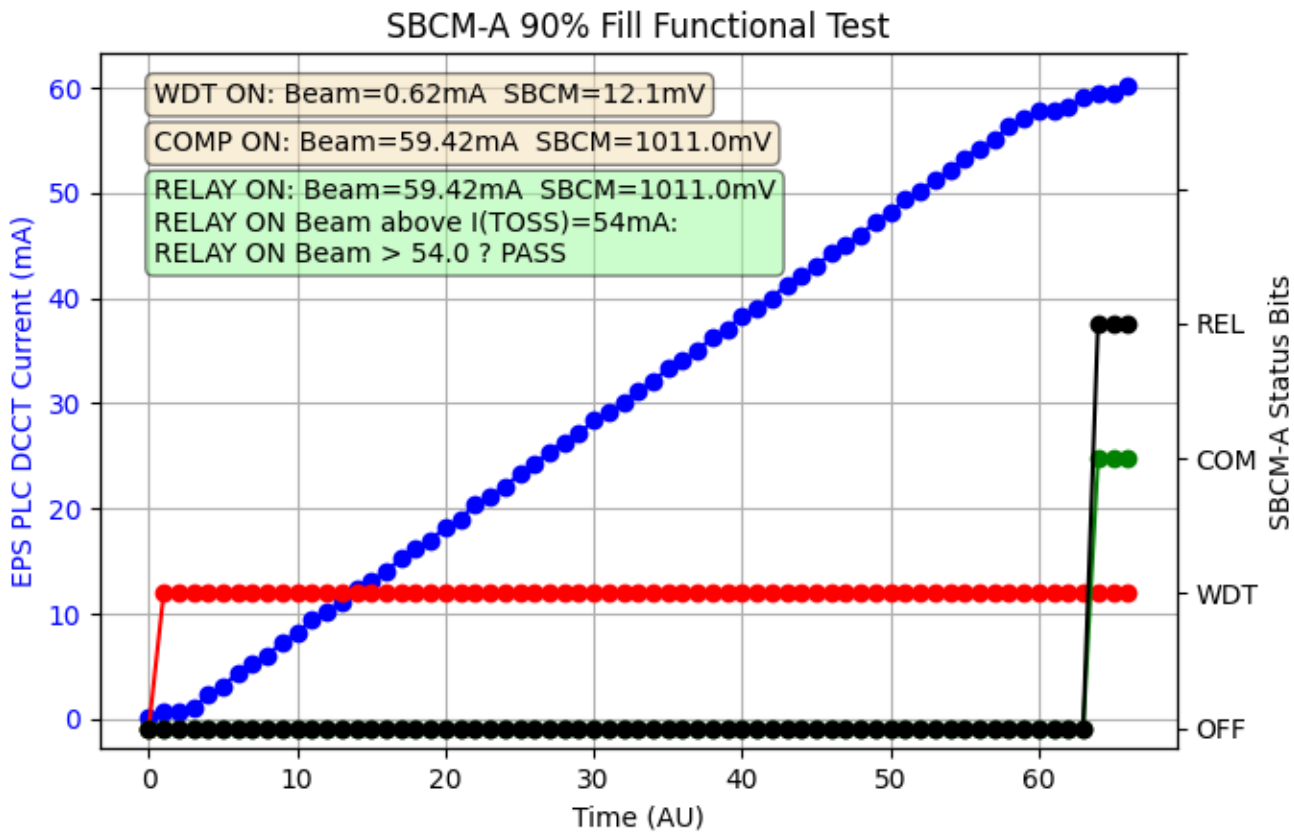
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

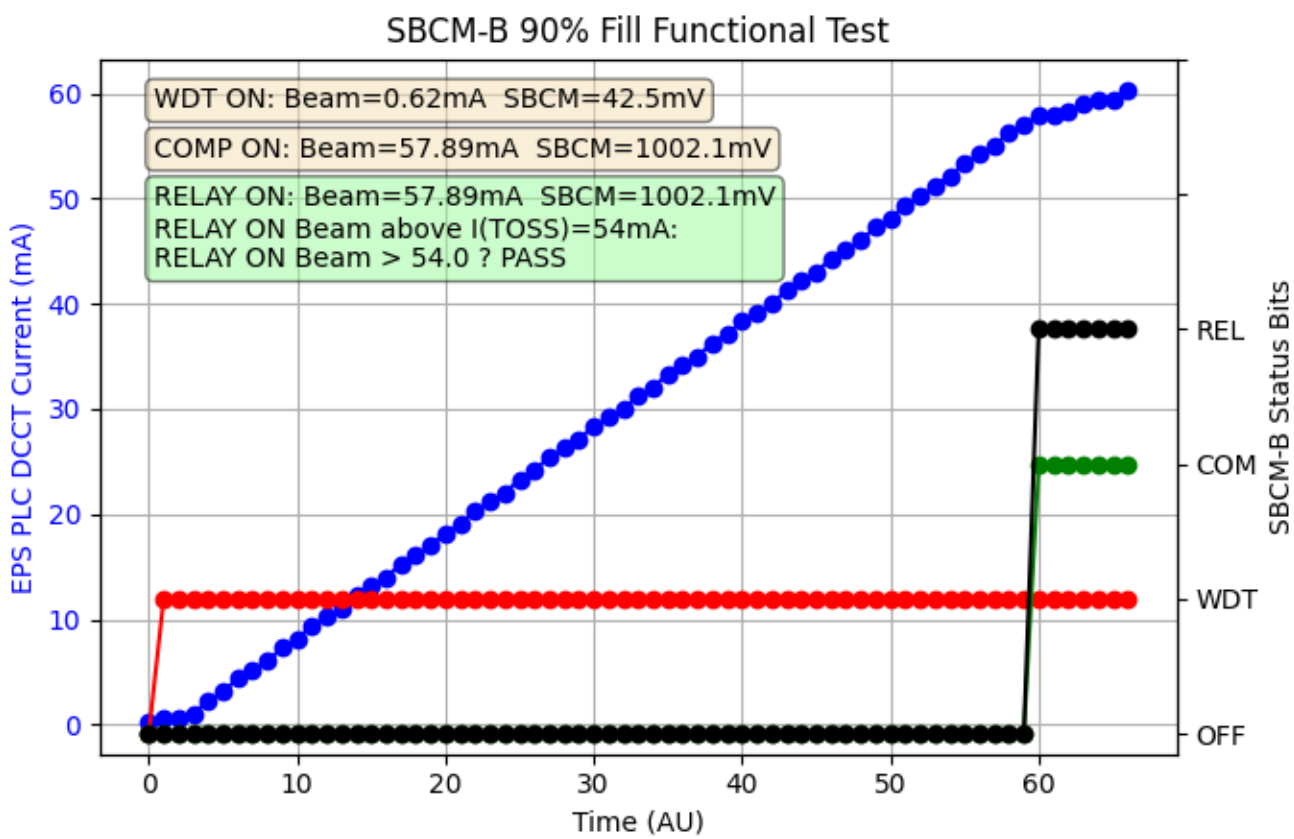
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC


SBCM-A 90% Fill Functional Test
WDT ON: Beam=0.62mA SBCM=12.1mV
COMP ON: Beam=59.42mA SBCM=1011.0mV
RELAY ON: Beam=59.42mA SBCM=1011.0mV
RELAY ON Beam above I(TOSS)=54mA:
RELAY ON Beam > 54.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-A Status Bits
REL
COM
WDT
OFF
Time (AU)


SBCM-B 90% Fill Functional Test
WDT ON: Beam=0.62mA SBCM=42.5mV
COMP ON: Beam=57.89mA SBCM=1002.1mV
RELAY ON: Beam=57.89mA SBCM=1002.1mV
RELAY ON Beam above I(TOSS)=54mA:
RELAY ON Beam > 54.0 ? PASS
EPS PLC DCCT Current (mA)
SBCM-B Status Bits
REL
COM
WDT
OFF
Time (AU)

SBCM Functional Scan with 90% Fill and Decreasing Beam

file: SBCM_FunctionalTestFalling_90pctFill_2025.txt

| Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R | Beam (mA) | Filt-A (mV) | Bits-A W,C,R | Filt-B (mV) | Bits-B W,C,R |
|---|---|---|---|---|---|---|---|---|---|
| 60.59 | 1031.83 | 1,2,3 | 1049.51 | 1,2,3 | 31.49 | 494.22 | 1,0,0 | 537.24 | 1,0,0 |
| 59.52 | 1012.47 | 1,2,3 | 1030.28 | 1,2,3 | 30.55 | 476.61 | 1,0,0 | 520.4 | 1,0,0 |
| 58.49 | 993.78 | 1,2,3 | 1012.56 | 1,2,3 | 29.56 | 458.31 | 1,0,0 | 502.92 | 1,0,0 |
| 57.86 | 982.57 | 1,0,0 | 1001.85 | 1,2,3 | 28.52 | 439.12 | 1,0,0 | 484.7 | 1,0,0 |
| 57.86 | 982.57 | 1,0,0 | 1001.85 | 1,2,3 | 27.46 | 418.31 | 1,0,0 | 464.7 | 1,0,0 |
| 57.59 | 977.44 | 1,0,0 | 996.95 | 1,0,0 | 26.39 | 398.96 | 1,0,0 | 445.78 | 1,0,0 |
| 57.59 | 977.44 | 1,0,0 | 996.95 | 1,0,0 | 25.39 | 380.65 | 1,0,0 | 428.05 | 1,0,0 |
| 57.52 | 975.75 | 1,0,0 | 995.26 | 1,0,0 | 24.4 | 362.54 | 1,0,0 | 411.16 | 1,0,0 |
| 56.56 | 957.94 | 1,0,0 | 978.47 | 1,0,0 | 23.5 | 345.75 | 1,0,0 | 395.1 | 1,0,0 |
| 55.56 | 939.82 | 1,0,0 | 961.24 | 1,0,0 | 22.53 | 328.98 | 1,0,0 | 378.96 | 1,0,0 |
| 54.53 | 920.46 | 1,0,0 | 942.87 | 1,0,0 | 21.53 | 310.25 | 1,0,0 | 360.94 | 1,0,0 |
| 53.56 | 903.22 | 1,0,0 | 926.54 | 1,0,0 | 20.4 | 289.12 | 1,0,0 | 340.66 | 1,0,0 |
| 52.49 | 883.6 | 1,0,0 | 907.96 | 1,0,0 | 19.5 | 272.46 | 1,0,0 | 324.63 | 1,0,0 |
| 51.56 | 865.04 | 1,0,0 | 890.3 | 1,0,0 | 18.4 | 253.28 | 1,0,0 | 306.25 | 1,0,0 |
| 50.56 | 847.26 | 1,0,0 | 873.22 | 1,0,0 | 17.57 | 238.62 | 1,0,0 | 292.06 | 1,0,0 |
| 49.53 | 828.76 | 1,0,0 | 855.53 | 1,0,0 | 16.44 | 218.47 | 1,0,0 | 272.07 | 1,0,0 |
| 48.57 | 810.88 | 1,0,0 | 838.62 | 1,0,0 | 15.44 | 200.08 | 1,0,0 | 254.6 | 1,0,0 |
| 47.5 | 791.03 | 1,0,0 | 820.04 | 1,0,0 | 14.51 | 183.14 | 1,0,0 | 238.09 | 1,0,0 |
| 46.57 | 773.16 | 1,0,0 | 803.16 | 1,0,0 | 13.54 | 167.11 | 1,0,0 | 222.26 | 1,0,0 |
| 45.54 | 753.53 | 1,0,0 | 784.93 | 1,0,0 | 12.54 | 148.91 | 1,0,0 | 204.63 | 1,0,0 |
| 44.54 | 735.56 | 1,0,0 | 767.47 | 1,0,0 | 11.54 | 131.61 | 1,0,0 | 187.75 | 1,0,0 |
| 43.54 | 716.35 | 1,0,0 | 749.63 | 1,0,0 | 10.51 | 112.96 | 1,0,0 | 169.5 | 1,0,0 |
| 42.54 | 697.95 | 1,0,0 | 732.01 | 1,0,0 | 9.58 | 96.29 | 1,0,0 | 152.93 | 1,0,0 |
| 41.51 | 678.72 | 1,0,0 | 713.64 | 1,0,0 | 8.55 | 78.92 | 1,0,0 | 135.65 | 1,0,0 |
| 40.54 | 660.62 | 1,0,0 | 696.22 | 1,0,0 | 7.58 | 63.23 | 1,0,0 | 119.77 | 1,0,0 |
| 39.54 | 642.92 | 1,0,0 | 679.14 | 1,0,0 | 6.55 | 46.12 | 1,0,0 | 101.91 | 1,0,0 |
| 38.58 | 624.83 | 1,0,0 | 662.11 | 1,0,0 | 5.58 | 30.96 | 1,0,0 | 85.18 | 1,0,0 |
| 37.55 | 605.82 | 1,0,0 | 643.95 | 1,0,0 | 4.52 | 18.21 | 1,0,0 | 67.74 | 1,0,0 |
| 36.58 | 588.05 | 1,0,0 | 626.95 | 1,0,0 | 3.62 | 11.97 | 1,0,0 | 54.1 | 0,0,0 |
| 35.58 | 569.31 | 1,0,0 | 609.12 | 1,0,0 | 3.55 | 11.53 | 1,0,0 | 52.81 | 0,0,0 |
| 34.55 | 550.52 | 1,0,0 | 591.24 | 1,0,0 | 3.09 | 9.86 | 0,0,0 | 46.85 | 0,0,0 |
| 33.59 | 532.92 | 1,0,0 | 574.29 | 1,0,0 | 2.56 | 8.46 | 0,0,0 | 41.46 | 0,0,0 |
| 32.55 | 513.66 | 1,0,0 | 555.84 | 1,0,0 | 1.56 | 7.28 | 0,0,0 | 36.08 | 0,0,0 |

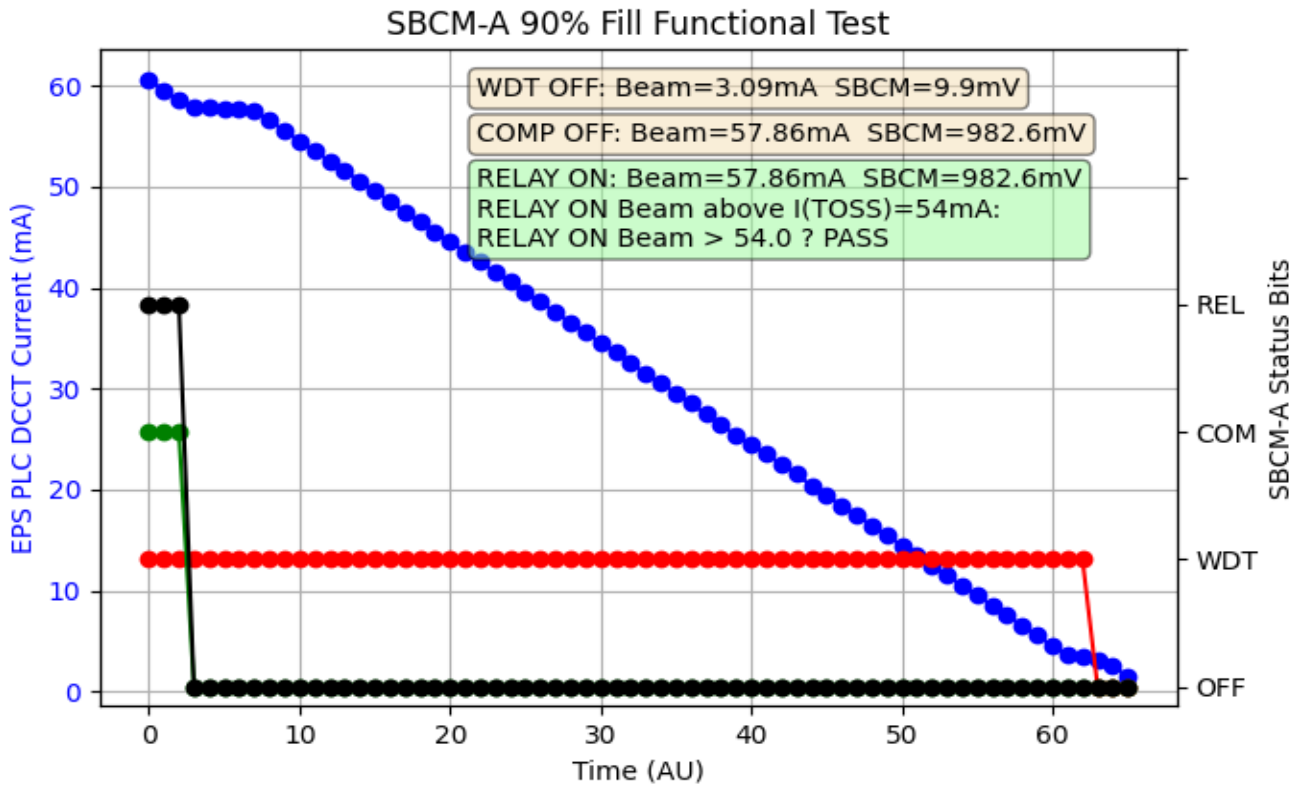
Beam: Storage Ring Beam as Measured by the EPS PLC Chassis in mA

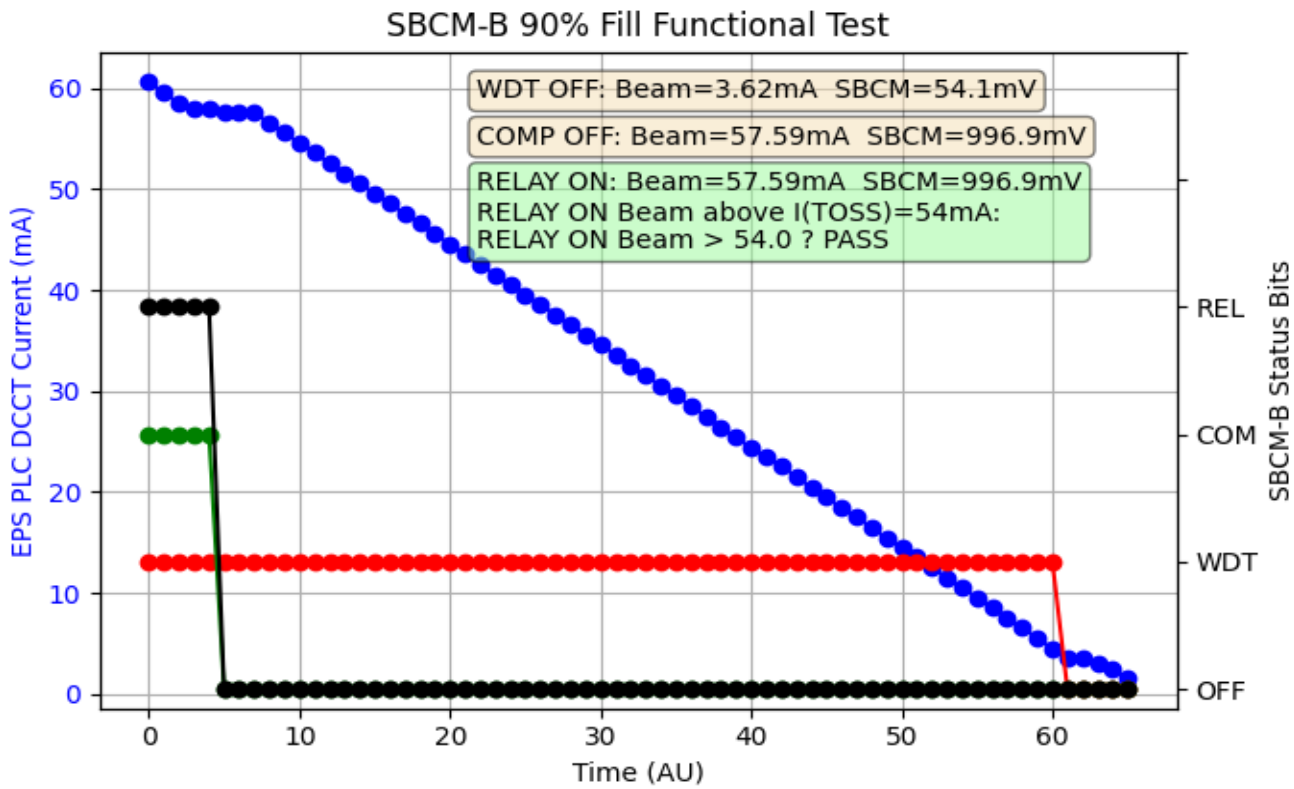
Filt-A: Chain-A Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-A: Chain-A State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC

Filt-B: Chain-B Rectifier Module Filter Output Measured by the Keithley 2100 in mV

Bits-B: Chain-B State of the WDT, Comparator, and Relay Status Bits Read From the SBCM PLC




| |
|---|
| All Tests for this Procedure have PASSED |
| TOSS SBCM Calibration Final Result: PASS |